Artificial Intelligence In Medicine 170 (2025) 103281

Contents lists available at ScienceDirect

Artificial Intelligence In Medicine

journal homepage: www.elsevier.com/locate/artmed

Preprocessing narrative texts in electronic medical records to identify hospital adverse events: A scoping review☆

Hamed Jafarpour [a,*], Guosong Wu [b], Cheligeer (Ken) Cheligeer [b], Jun Yan [a], Yuan Xu [b], Danielle A. Southern [b], Cathy A. Eastwood [b], Yong Zeng [a], Hude Quan [b,*]

[a] Concordia University, Gina Cody School of Engineering and Computer Science, Concordia Institute for Information Systems Engineering, 1515 Sainte Catherine West, Montreal, H3G 2W1, Quebec, Canada
[b] University of Calgary, Department of Community Health Sciences, Cumming School of Medicine, 2500 University Drive NW, Calgary, T2N 1N4, Alberta, Canada

ARTICLE INFO

Keywords:
Preprocessing
Narrative EMR
Hospital adverse event
Natural language processing
Large language model
Clinical text

ABSTRACT

Background: Narrative electronic medical records (EMR), which include textual notes created by clinicians within healthcare environments, represent a significant resource for documenting various facets of patient care. This form of text exhibits distinctive characteristics, such as the occurrence of grammatically incorrect sentences, abbreviations, frequent acronyms, specialized characters with particular meanings, negation expressions, and sporadic misspellings. As a result, a primary goal in processing these textual notes is to implement effective preprocessing techniques that enhance data quality and ensure consistency across all entries. Recent advancements in algorithms and methodologies within the fields of natural language processing (NLP), machine learning (ML), and large language models (LLM) have prompted researchers to leverage narrative EMR for the detection of hospital adverse events (HAE).

Methods: The scoping review adhered to the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews) guidelines. A scoping review protocol was developed and utilized to guide the research process, clearly outlining the eligibility criteria, information sources, search strategies, data management, selection process, data collection procedures, data items, outcomes and prioritization, data synthesis, and meta-bias considerations. The search strategy was implemented across nine engineering and medical electronic databases.

Results: The results have indicated that from a total of 3,264 studies retrieved, 48 unique studies were included in the review. Responses to the research questions were systematically extracted from these studies. The review has identified challenges associated with the preprocessing of narrative texts in EMR for HAE identification. Additionally, three research gaps have been identified: (1) the imperative need for a pipeline to preprocess narrative EMR for the identification of HAE, (2) the necessity for a robust system capable of managing the extensive volume of narrative EMR data, and (3) the requirement for temporal event system, which are essential for effective HAE detection. The study also has underscored the essential role of preprocessing tasks in enhancing the performance of HAE detection. The study has emphasized the importance of extracting N-grams from clinical text, normalizing these N-grams through lemmatization and/or stemming, and establishing semantic feature extraction in preprocessing tasks that significantly affect HAE detection performance. While LLM-based systems naturally incorporate tokenization and normalization processes within their frameworks, it remains crucial to address features that hold semantic relevance to the specific type of HAE during preprocessing.

Conclusion: This scoping review has provided valuable insights for researchers focused on HAE detection utilizing narrative EMR data. It has elucidated how preprocessing tasks can elevate the performance of HAE

☆ This scoping review is funded by the Canadian Institutes of Health Research (CIHR), Canada.
* Corresponding authors.
E-mail addresses: hamed.jafarpour@concordia.ca (H. Jafarpour), Guosong.wu@ucalgary.ca (G. Wu), cheligeerken@ucalgary.ca (C. Cheligeer), jun.yan@concordia.ca (J. Yan), yuxu@ucalgary.ca (Y. Xu), dasouthe@ucalgary.ca (D.A. Southern), caeastwo@ucalgary.ca (C.A. Eastwood), yong.zeng@concordia.ca (Y. Zeng), hquan@ucalgary.ca (H. Quan).

https://doi.org/10.1016/j.artmed.2025.103281
Received 6 December 2024; Received in revised form 24 September 2025; Accepted 5 October 2025
Available online 8 October 2025
0933-3657/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

detection and draws attention to neglected research gaps within the field. Addressing these gaps will necessitate further investigation in subsequent research endeavors.

Contents

1. Introduction ... 2
   1.1. Rationale ... 2
   1.2. Objectives ... 3
   1.3. Statement of significance ... 3
2. Methods ... 3
   2.1. Study design ... 3
   2.2. Information sources ... 3
   2.3. Search strategy ... 4
   2.4. Eligibility criteria ... 4
   2.5. Selection process ... 4
3. Results ... 4
   3.1. Search result ... 4
   3.2. Preprocessing techniques for handling large volumes of narrative EMR data (RQ1) ... 5
   3.3. Preprocessing techniques to normalize text across narrative EMR data (RQ2) ... 5
   3.4. Application of feature engineering techniques in the preprocessing narrative EMR data (RQ3) ... 5
   3.5. Synthesis of results ... 5
4. Discussion ... 7
   4.1. Challenges in preprocessing narrative EMR data to identify HAE (RQ4) ... 8
   4.2. Outcome ... 8
      4.2.1. Primary outcome ... 8
      4.2.2. Secondary outcome ... 10
   4.3. Major findings and future works ... 14
   4.4. Limitations of this work ... 15
5. Conclusion ... 16

CRediT authorship contribution statement ... 16
Declaration of competing interest ... 16
Acknowledgments ... 16
Appendix A. PRISMA checklist ... 16
Appendix B. Study protocol ... 16
Appendix C. Supplementary data ... 16
References ... 16

1. Introduction

1.1. Rationale

Hospital adverse events (HAE) refer to unintended injuries or complications that occur during a patient's hospitalization, resulting in prolonged stays, disability upon discharge, or even mortality. These events are primarily related to the management of healthcare rather than pre-existing medical conditions of the patient [1]. In addition to the direct consequences for patients, HAE impose a significant financial burden on the healthcare system [2,3]. According to the economic patient safety report published in 2017, the most financially burdensome HAE include healthcare-associated infections (HAI), Venous thromboembolism (VTE), pressure ulcers, medication errors and incorrect or delayed diagnoses [2]. This critical issue has led researchers to investigate HAE, leading to the publication of systematic reviews (e.g., [1,4]) and scoping reviews (e.g., [5,6]) focused on the detection of HAE.

Advancements in algorithms and methodologies within the domains of natural language processing (NLP), machine learning (ML), and large language models (LLM) have encouraged researchers to use narrative EMR for the identification of HAE. Narrative EMR, which encompasses textual notes generated by clinicians in healthcare settings, represents unstructured data that serves as a valuable resource, thoroughly documenting various aspects of patient care. This type of text possesses unique characteristics, including the presence of grammatically incorrect sentences and phrases, abbreviated terms, frequent use of acronyms, specialized characters with specific meanings, negation expressions, and occasional misspellings [7]. Consequently, a primary objective in processing these textual notes is to implement effective preprocessing techniques. This process aims to enhance data quality and ensure consistency across all textual entries, thereby transforming them from an unstructured format into a semi-structured representation of clinical information.

The preprocessing of narrative EMR constitutes a multifaceted process that encompasses three primary tasks: managing substantial volumes of narrative EMR data, text normalization, and feature engineering. The management of large volumes of narrative EMR data necessitates the implementation of effective preprocessing techniques that facilitate the organization and segmentation of extensive clinical text, thereby ensuring the preservation of critical information for subsequent analysis. This step is crucial for optimizing the input size for ML and LLMs, ultimately enhancing their capacity to accurately identify HAE.

Text normalization represents a vital preprocessing activity that strives to achieve uniformity across all terms within the clinical text, converting it into a standardized and noise-free format. This task involves word tokenization and normalization. Word tokenization entails the identification of individual words (or phrases), abbreviations, and special characters, while word normalization focuses on interpreting clinical abbreviations and special characters, rectifying misspelled words, and establishing uniformity among similar terms within the clinical text. For instance, the sentence "Pt is 65 yo m w/ -fx referred to pt for lbp" is transformed through the processes of text normalization to read as "Patient is 65 years old male with no fracture referred to physical therapy for low back pain".

Feature engineering incorporates information extraction (IE) methods and numerical representation techniques, commonly referred to as vectorization. The NLP approach employs syntactic, semantic, and

sentiment analyses, which represent the three fundamental forms of analysis utilized to extract information effectively. In the aforementioned example, the extracted information would yield: "age: 65, sex: male, medical diagnosis: low back pain, no fracture, treatment: physical therapy".

The motivation for undertaking this scoping review arises from the following inquiries.

- Why using narrative EMR is essential for detecting HAE? Generally, HAE detection has relied on methods such as manual chart reviews, discharge diagnostic codes, prevalence surveys, and incident reporting systems [8]. However, manual chart reviews are often characterized by their time-consuming, resource-intensive, and costly nature [9,10]. Additionally, discharge diagnostic codes exhibit low sensitivity and positive predictive value (PPV) in accurately identifying HAE cases [11]. Similarly, incident reporting systems tend to underestimate the true incidence rate of HAE [12]. Prevalence surveys are also recognized for their inefficiency, lack of scalability, and variations in inter-observer assessments regarding HAE detection [10,13]. The diverse coding methodologies employed by various institutions further complicate benchmarking efforts [11,14]. Given these limitations and the advancements in NLP, artificial intelligence (AI) and in particular LLMs techniques, there is a growing imperative among researchers to adopt narrative EMR for the detection of HAE. Moreover, Bates et al. undertook a scoping review that elucidates the potential role of AI and the exploitation of unstructured data in identifying HAE [3].
- Why is preprocessing narrative EMR data essential? As previously noted, narrative EMR data constitutes a valuable form of unstructured data; however, due to the inherent characteristics of this textual note, preprocessing is a critical initial step when utilizing the data for any medical applications, particularly in the context of HAE detection. Additionally, several scoping reviews have highlighted the challenge posed by the limited availability of high-quality data within textual notes, which contributes to inconsistencies in the detection of HAE [6,15,16]. Consequently, the effective preprocessing of narrative EMR data is essential in order to produce high-quality data, thereby ensuring the performance and reliability of HAE detection initiatives.
- Why is preprocessing narrative EMR data still essential with the advent of LLMs? The preprocessing narrative EMR data remains an essential component in the context of advancing LLMs, driven by several significant factors. Narrative EMR data often consists of extensive textual notes that may exceed the input token limitations of various language models. For instance, the BERT model has a maximum token capacity of 512 tokens in its large variant, as does T5. The GPT-2 model accommodates a maximum of 1024 tokens, while GPT-3 supports up to 4096 tokens. In contrast, both GPT-4 and LLAMA 3 have a token limit of 8192.
  Furthermore, the architecture of LLMs is predicated on precise tokenization as an initial processing phase. Given the complexity inherent in medical terminology, meticulous normalization is imperative to guarantee that LLMs accurately interpret specialized language. Therefore, preprocessing is vital for sustaining the quality and reliability of predictions, particularly in tasks such as HAE detection, as it addresses input size limitations, aligns the data with model architecture, and ensures consistent interpretation of medical information.

Acknowledging the considerable challenges and opportunities associated with the utilization of narrative EMR data for the detection of HAE, as well as the essential requirement for preprocessing, served as the impetus for conducting this scoping review.

1.2. Objectives

Scoping reviews are typically conducted for a variety of purposes, primarily to examine the breadth and depth of existing literature, organize and succinctly summarize available evidence, guide future research initiatives, and identify research gaps [17,18].

The objective of this scoping review is to provide an overview of the preprocessing of narrative EMR data, aiming to summarize current research and identify underexplored areas in the preprocessing of narrative EMR data for the detection of HAE. The Joanna Briggs Institute (JBI) recommends the utilization of the population, concept, and context (PCC) framework for scoping reviews [17]. Consequently, the PCC model serves as a foundational step in guiding the formulation of the research questions and the inclusion of relevant information in this review. Accordingly, the PCC model is delineated as follows:

- Population: Researchers, practitioners, and developers working with narrative EMR.
- Concept: Preprocessing narrative EMR.
- Context: HAE detection using narrative EMR data with NLP and ML techniques.

In accordance with the outlined PCC, this study seeks to address the following research questions:

- Research question 1 (RQ1): What preprocessing techniques are recognized for managing large volumes of narrative EMR data to identify HAE?
- Research question 2 (RQ2): What is known about preprocessing narrative EMR data to normalize it for identifying HAE?
- Research question 3 (RQ3): What is known about the application of feature engineering techniques in the preprocessing narrative EMR data to detect HAE?
- Research question 4 (RQ4): What are the main challenges in the preprocessing narrative EMR data that hinder the effectiveness of HAE detection?

To elucidate the rationale behind these research questions, it is essential first to define the overarching concept of preprocessing within the realm of general text analysis. In general, text preprocessing refers to the initial tasks taken to normalize raw text data for further analysis or modeling. Text preprocessing encompasses three primary tasks: sentence segmentation, word tokenization, and word normalization [19]. The narrative text in EMR data comprises a substantial volume of information from diverse healthcare settings. In the manual process of detecting HAE, the chart reviewers handle the large volume of text by attending to each section to specify its context (e.g., "past medical history" or "present illness"). The chart reviewers compile the comprehensive medical history of the patient from various healthcare settings. If an event is recorded in the past medical history, indicating potential adverse events, it is not classified as a HAE if it did not occur during a hospital stay. Subsequently, they normalize the concept and finally extract the information related to HAE, and then make a decision about whether HAE is happening or not. In accordance with the scenario presented and the primary concept of preprocessing as defined by Jurafsky in [19], this preprocessing procedure consists of three principal tasks: managing a substantial volume of data, text normalization, and feature engineering. As a result, the research question emphasizes the management of the significant data volume as the initial focus, followed by the subsequent processes of text normalization and feature engineering.

1.3. Statement of significance

This scoping review aims to clarify the significance of preprocessing narrative EMR in the identification of HAE and to highlight the existing research gaps in this area.

- **Problem:** The preprocessing of narrative EMR data is critical for the effective detection of HAE. However, the current preprocessing pipelines have not been specifically tailored to meet the requirements for HAE detection.
- **What is already known:** Researchers have acknowledged the substantial value of narrative EMR data for the detection of HAE [3]. Furthermore, they have identified that preprocessing is a critical component in this context. Although preprocessing techniques have been implemented in various research initiatives, the specific tasks that significantly enhance HAE detection remain to be determined.
- **What this paper adds:** This study aims to present a comprehensive overview of the preprocessing of narrative EMR data for the detection of HAE. It will summarize the preprocessing tasks described in existing research and identify gaps that have been overlooked. Furthermore, the study will elucidate which specific preprocessing tasks contribute significantly to enhancing the performance of HAE detection.

The subsequent sections of this paper are organized as follows. Section 2 outlines the method employed for the scoping review, detailing the study design, information sources, search strategy, eligibility criteria, and the process of paper selection. Section 3 presents the results of the search, addresses the research questions, and synthesizes the results. Section 4 provides a discussion that encompasses the challenges associated with preprocessing narrative texts in EMR, identifies research gaps that have been overlooked in this area, and examines the influence of preprocessing tasks on the effectiveness of HAE detection. This section also emphasizes the key findings, suggests directions for future research, and acknowledges the limitations of the current scoping review. Finally, Section 5 concludes the review.

2. Methods

2.1. Study design

The research methodology employed in this scoping review adheres to the guidelines set forth in the JBI Manual for Evidence Synthesis [20]. The PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews) guidelines [21] were meticulously followed to ensure a transparent review process. The PRISMA checklist is presented in Appendix A.

A protocol for the scoping review was developed and implemented to direct the research process. The steps outlined in this protocol are in accordance with the PRISMA-ScR guidelines [21]. In line with the findings of Tricco et al. [21], it is essential that the protocol checklist for systematic reviews, as detailed in [22], is modified. Specifically, two items — namely "risk of bias in individual studies" and "confidence in cumulative evidence" — should be excluded from the scoping review protocol checklist. The scoping review protocol along with the corresponding checklist can be found in Appendix B.

Given the significance of information sources, the search strategy, eligibility criteria and selection process are succinctly outlined in this section.

2.2. Information sources

To undertake this review, nine sources of medical and engineering data were utilized, which include PubMed, Cochrane RCT, ISI Web of Science, EBSCO (Library, Information Science & Technology Abstracts), Ovid Journals, ACM Digital Library, Scopus, and IEEE. Additionally, Google Scholar was employed to identify relevant articles published as multidisciplinary studies. Furthermore, a search of gray literature was conducted through OpenGrey.

Table 1
Inclusion and exclusion criteria.

Criteria	Description
Inclusion (I1)	The studies employed clinical narratives as a means to identify HAE.
Inclusion (I2)	Research studies that were published subsequent to January 1, 2005.
Inclusion (I3)	The clinical textual notes has been composed in the English language.
Inclusion (I4)	The articles that have been published are written in the English language.
Exclusion (E1)	The studies published prior to January 1, 2005.
Exclusion (E2)	The studies employed exclusively structured data as the input for HAE detection.
Exclusion (E3)	The studies conducted were systematic or scoping reviews.
Exclusion (E4)	The studies were not available for unrestricted access to the authors.

2.3. Search strategy

Initially, we conducted a search across three databases: Prospero [23], OSF [24], and Figshare [25]. The search terms utilized included "Preprocessing narrative electronic medical record", "Preprocessing of clinical narratives", "Preprocessing of textual notes", and "Preprocessing of clinical free-text notes" in an effort to identify any ongoing or planned reviews. Unfortunately, this initial search yielded no relevant results.

Subsequently, we advanced to the second phase, which entailed formulating search terms pertinent to the scoping review topic. We experimented with multiple variations of these terms to optimize the balance between retrieving relevant articles and managing the overall volume of results. Ultimately, we identified keywords that aligned with the research questions of the scoping review, including "preprocessing", "natural language processing", "machine learning", "large language model", "narrative electronic medical record", "adverse event", and "hospital adverse event". Additionally, synonymous concepts and terms closely related to the selected keywords were integrated using Boolean operators such as "AND" and "OR" to enhance the efficiency of the search outcomes. This comprehensive search string was subsequently employed in queries across nine information sources. Further details are available in the search strategy section, which can be found in Appendix B.

2.4. Eligibility criteria

The study's inclusion and exclusion criteria are outlined in Table 1. Criterion I1 allows for the inclusion of articles using clinical narratives and those combining clinical narrative with structured data. The focus is on articles published since January 1, 2005 (i.e. I2), covering the first paper on automatic HAE detection via NLP by Melton [26]. Criterion I3 assesses the preprocessing of English clinical text, considering the distinct needs of different languages, while Criterion I4 addresses the costs of translating non-English articles. In conjunction with the inclusion criteria, four specific exclusion criteria were outlined (i.e. E1–E4). The review includes published articles from conferences and journals across nine databases in the medical and engineering fields, ensuring a comprehensive representation of relevant literature.

2.5. Selection process

The selection process for the scoping review involved two independent reviewers who screened abstracts and titles against predetermined eligibility criteria. Studies approved by both reviewers proceeded to full-text evaluation, with agreement among reviewers assessed utilizing Kappa statistics [27]. Any discrepancies were resolved through consensus, with the option to consult a third expert reviewer when necessary.

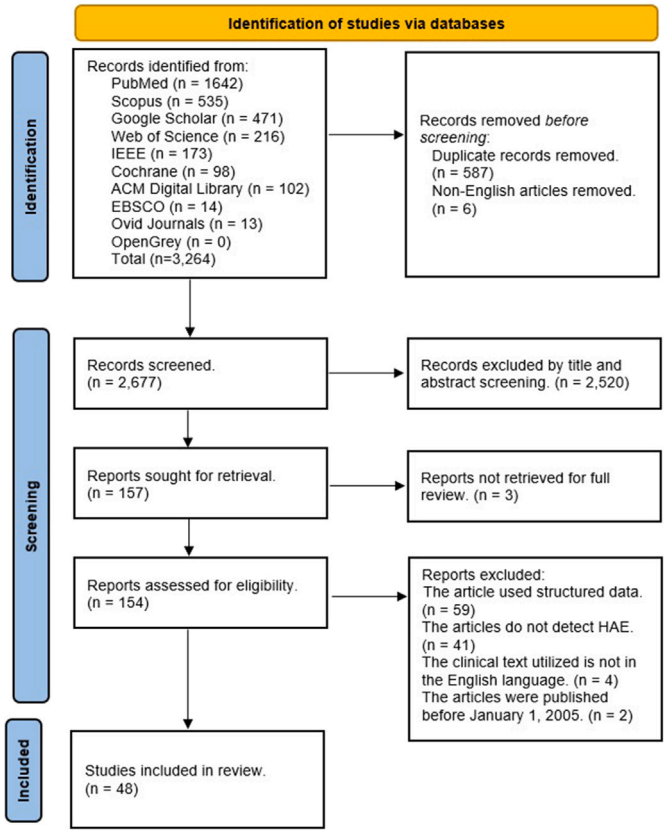


Fig. 1. PRISMA flow diagram of the source selection process.

Additionally, two supplementary expert reviewers — one specializing in NLP and ML (J.Y.) and another focusing on HAE (Y.X.) — evaluated the qualifying studies. Only those papers achieving consensus among all four reviewers were included in the full-text review.

3. Results

3.1. Search result

The final search for relevant articles was conducted on May 27, 2025, yielding 3264 results. After removing 587 duplicates and applying inclusion and exclusion criteria, 6 non-English articles (related to I4 criterion), 2 pre-2005 articles (related to E1 criterion), 100 unrelated articles (related to E2 criterion), 49 review articles (related to E3 criterion), and 3 inaccessible articles (related to E4 criterion) were excluded . Ultimately, 48 articles were selected for full-text review and data extraction. Fig. 1 depicts the PRISMA flow diagram of the source selection process. Details of the search results from each data source are provided in the figure, along with a clear representation of the selection process based on inclusion and exclusion criteria.

Among the 48 articles selected for full-text review, 72% were published after the year 2020, while 28% were published between 2005 and 2020. The most recent update of the review was conducted on May 27, 2025. Over the course of five months in 2025, eight articles have been published, marking the highest annual publication total to date and constituting 16% of the overall output. Furthermore, 75% of the papers received funding support. The majority of the publications originated from the United States, representing 56% of the total, followed by Canada, which contributed 10% of the publications. The remaining articles were disseminated from China, Spain, the United Kingdom, Brazil, Australia, Italy, the Netherlands, Denmark, France, and Korea. Fig. 2 illustrates that the research articles using narrative EMR for

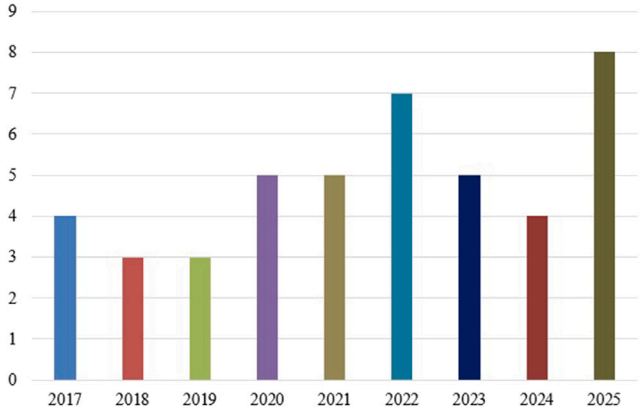


Fig. 2. Annual publications on using narrative EMR to detect HAE.

detecting HAE have significantly increased in recent years. Initially, only a few publications appeared in 2005, 2011, 2013, and 2015. A notable rise began in 2017, with four publications that year, increasing to seven in 2022 and eight in the first five months of 2025. This trend highlights growing interest and progress in the field, emphasizing the importance of narrative EMR in identifying HAE. Annual publication counts from 2017 to 2025 reflect this significant uptick starting in 2017.

Across the reviewed studies, VTE and its subtypes — deep vein thrombosis (DVT) and pulmonary embolism (PE) — were the most frequently targeted hospital adverse events, appearing in [26,28–42]. Several studies addressed postoperative complications in general, such as [43–51], while others focused on specific infection types including surgical site infections (SSI) [52–54], or intra-abdominal SSI [55], or reoperation for wound infection [56]. Additional HAE categories included falls [26,57–59], adverse drug reactions [60,61], immune-related adverse events [62,63], sepsis [35,43,64,65], anesthesia-related adverse events [66], and other targeted events such as hysteroscopic sterilization device removal [67], peripheral intravenous infiltrations and extravasations [68], and acute renal failure requiring dialysis [35].

3.2. Preprocessing techniques for handling large volumes of narrative EMR data (RQ1)

In hospitals, a significant amount of narrative EMR is generated through various reports, such as radiology and nursing reports, when patients come for treatment. Although NLP, ML techniques, and LLMs are effective for HAE detection, they need preprocessed data to detect HAE effectively. Section detection is crucial in preprocessing, as it identifies portions of clinical text that contain relevant information regarding a patient. Not all sections are useful for HAE detection, making the selection of pertinent sections essential. Research studies have employed two techniques to effectively reduce the substantial volume of narrative EMR data, which includes text and sentence segmentation as section detection, and subsequently select the pertinent section. The techniques utilized in these studies are summarized in Table 2. Furthermore, the types of datasets utilized in each study are summarized in Table 2. The entries delineate whether the data source is classified as public, private, or a combination of both public and private datasets. This classification offers valuable context regarding data accessibility and the generalizability of the findings.

3.3. Preprocessing techniques to normalize text across narrative EMR data (RQ2)

Text normalization in narrative EMR data involves converting it into a consistent and noise-free format through two main components:

Table 2
Section detection and text normalization techniques.

Ref.	Dataset	Section detection[a]	Word tokenization[b]	Word normalization[c]
[26]	Private	SS	POS, N-grams	Abbr, Synonyms
[28]	Private	–	N-grams	Lemma
[29]	Private	–	–	–
[30]	Private	SS	–	–
[31]	Private	TS, SS	SeChr, POS, N-grams	Synonyms
[32]	Private	TS, SS	POS, N-grams	Abbr
[33]	Private	–	SeChr, N-grams	CF, Lemma
[34]	Private	TS	–	Abbr, Synonyms
[35]	Private	SS	–	–
[36]	Private	TS	–	–
[37]	Private	–	–	–
[38]	Private	–	–	CF, Synonyms
[39]	Private	–	–	Synonyms
[40]	Private	–	–	Synonyms
[41]	Private	–	–	Synonyms
[42]	Private	SS	POS, N-grams	–
[43]	Private	–	SeChr	Abbr
[44]	Private	SS	SeChr, N-grams	CF, Lemma
[45]	Private	–	–	–
[46]	Private	–	–	–
[47]	Private	–	–	–
[48]	Private	–	SeChr	Lemma
[49]	Public and private	–	–	–
[50]	Private	–	–	–
[51]	Public	–	–	Lemma
[52]	Private	TS, SS	SeChr, POS, N-grams	CF, Spell checking
[53]	Private	–	N-grams	Lemma
[54]	Private	TS	SeChr	CF
[55]	Public	–	N-grams	–
[56]	Private	–	SeChr	CF, Lemma
[57]	Private	TS, SS	SeChr, POS, N-grams	CF, Lemma
[58]	Private	–	SeChr	CF
[59]	Private	–	–	–
[60]	Private	–	–	–
[61]	Private	TS	SeChr, Clitic	CF, Abbr
[62]	Private	TS, SS	SeChr, POS, N-grams	–
[63]	Private	TS, SS	–	–
[64]	Private	SS	N-grams	–
[65]	Private	–	–	–
[66]	Private	–	SeChr, N-grams	Lemma
[67]	Public	TS, SS	–	–
[68]	Private	–	N-grams	Spell checking, Synonyms
[69]	Public and private	TS, SS	SeChr, POS, N-grams	CF
[70]	Private	–	SeChr, N-grams	CF
[71]	Private	–	N-grams	Synonyms
[72]	Private	–	SeChr	Lemma
[73]	Public	SS	SeChr, POS, N-grams	Synonyms
[74]	Private	–	–	CF, Lemma

[a] "TS" = "Text segmentation", "SS" = "Sentence segmentation".
[b] "SeChr" = "Special characters".
[c] "CF" = "Case folding", "Abbr" = "Abbreviation", "Lemma" = "Lemmatization and/or stemming".

tokenization and normalization. Tokenization refers to the process of dividing text into discrete units, or tokens, which may include words, abbreviations, and special characters. Key techniques include detecting N-grams, handling special characters, Part-Of-Speech (POS) tagging, and separating clitics. Normalization is a process that standardizes tokens to promote consistency. The techniques to handle text normalization include deciphering abbreviations, fixing misspellings, addressing synonyms, case folding, and lemmatization/stemming. Tokenization and normalization are essential components of preprocessing EMR data, particularly for the accurate identification of HAE. These processes enhance data clarity, consistency, and quality, thereby improving the effectiveness of NLP techniques, ML models, and LLMs.

The concepts of tokenization and normalization techniques are presented in the fourth and fifth columns of Table 2, respectively.

3.4. Application of feature engineering techniques in the preprocessing narrative EMR data (RQ3)

Feature engineering for textual data encompasses IE methodologies, such as the identification of words and phrases, as well as vectorization techniques [75].

To date, several reviews have been conducted regarding IE from medical text. Wang et al. reviewed clinical IE applications, categorizing them into rule-based and ML approaches, highlighting uses like smoking status identification and drug extraction [76]. Navarro et al. focused on clinical named entity recognition (NER) and relation extraction, finding that the most common entities extracted were "problem", "test" and "treatment", along with various medical details [77]. Fu et al. reviewed clinical concept extraction, classifying features into four categories: linguistic, domain knowledge, statistical, and general document features [78].

The distinction between the IE approach utilized in the current study and prior research lies in the present study's emphasis on linguistic features relevant to IE, as well as its focus on obtaining precise temporal information associated with the extracted information.

Linguistic features relevant to IE are classified into several categories [79]: the removal of stop words, recognition of named entities, keyword searching, chunking,[1] POS tagging, extraction of N-grams through shallow parsing, elimination of infrequent words, filtering of section headers, and management of negation terms. The handling of negation encompasses both explicit negation (e.g., "no" or "not") and implicit negation (e.g., "deny" or "absent"). Linguistic features relevant to IE employed in the analysis of full-text articles are outlined in Table 3.

Temporal information pertaining to the extracted data is essential for determining the timing of events—specifically whether they occurred in the past, are currently in progress, or fall within a specified timeframe following discharge (for example, 30 to 40 days). The integration of this temporal data is crucial for ensuring the accuracy of HAE detection. Among the various tools mentioned in Table 3, MedLee [80], PyConText [81], IDEAL-X [32], and EasyCIE-PEDVT [31] incorporate features that capture timestamp.

3.5. Synthesis of results

This study has thus far addressed the first three research questions related to the preprocessing of narrative EMR data with the aim of identifying HAE. By analyzing the data through various dimensions of preprocessing, including handling large volumes of narrative EMR data (i.e., RQ1), text normalization techniques (i.e., RQ2), and feature engineering strategies (i.e., RQ3), it is possible to discern overarching patterns, relationships, and implications that extend beyond the findings presented in individual sections.

- The findings presented in Tables 2 and 3 indicate that, in order to manage the substantial volume of narrative EMR data effectively, three methodologies have been implemented: (1) section detection, (2) targeted keyword searches pertaining to HAE, and (3) the application of prompting and chunking techniques. Certain studies have employed text and sentence segmentation to effectively manage large amounts of data (e.g., [31,32,52,57,67,69]). Additionally, there are investigations that focus on conducting keyword searches related to HAE. These searches may be guided either by healthcare professionals who select keywords (e.g., [28,29,43,55], and [60]) or by utilizing a pre-established clinical terminology dataset (e.g., [37,58,68]). Furthermore, with the advancements in LLMs, recent studies have implemented prompting and chunking techniques as a means to handle substantial narrative EMR data volumes (e.g., [42,49,64], and [65]).
- Word tokenization and normalization, which represent two essential components of text normalization, are commonly employed in studies utilizing rule-based and ML algorithms (e.g., [26,31,32,52,57,67,69]). However, recent research involving LLMs, which are constructed upon the transformer architecture and inherently incorporate tokenization and embedding layers, has not necessitated the use of text normalization. For instance, studies such as [42,45,46,49,64], and [65] illustrate this trend.
- The findings indicate that studies employing rule-based and machine learning approaches to detect HAE predominantly focused on searching for keywords associated with HAE, utilizing named entity recognition, and addressing negation concepts to extract relevant information (e.g., [26,31,32,35–37,57,58,67–69]). In contrast, studies utilizing LLMs implemented techniques such as prompting, chunking, and the Chain-of-Thought (COT) method for information extraction (e.g., [42,45,46,49,64,65]). Furthermore, because vectorization techniques are inherently integrated within the architecture of LLMs, these studies did not apply alternative vectorization methods.

In summary, the synthesis of findings reveals a clear evolution in methodologies for processing narrative EMR data. Traditional approaches, primarily based on rule-based and ML techniques, relied heavily on keyword searches, named entity recognition, and negation detection, often requiring extensive text normalization. In contrast, recent advancements leveraging LLMs have introduced more sophisticated strategies, including prompting, chunking, and COT reasoning, to manage and extract information from large-scale unstructured data. Notably, the use of LLMs has obviated the need for explicit text normalization and vectorization, as the models' transformer-based architectures inherently handle these processes. This transition underscores a shift toward more integrated and scalable solutions for preprocessing narrative EMR data.

4. Discussion

This discussion section reflects on the principal findings of the study by examining the challenges associated with preprocessing narrative EMR data, evaluating both primary and secondary outcomes, and critically assessing the strengths and limitations of this review. Initially, the challenges related to EMR preprocessing, as addressed in RQ4, will be examined, with particular attention devoted to managing the substantial volume of narrative EMR data, text normalization, and the implementation of feature engineering techniques. Subsequently, the discussion will focus on the overlooked research gap in this field and the evaluation of the effectiveness of HAE detection as influenced by various preprocessing tasks, which constitute the outcomes of interest. Finally, this section will offer a comprehensive reflection on the strengths and limitations of the present work, while also delineating potential directions for future research.

4.1. Challenges in preprocessing narrative EMR data to identify HAE (RQ4)

This section delineates the significant challenges associated with preprocessing narrative EMR data that impede the detection of HAE. These challenges are categorized into three primary areas: managing the substantial volume of narrative EMR data, text normalization, and the application of feature engineering techniques. Table 4 provides an overview of these categorized challenges along with their corresponding limitations, which affect HAE detection.

4.2. Outcome

The primary objective of this scoping review is to identify overlooked research gaps in the preprocessing of narrative EMR data for the detection of HAE utilizing NLP and ML models. The secondary objective is to address the established research questions by synthesizing responses and evaluating the effectiveness of HAE detection as influenced by various preprocessing tasks.

4.2.1. Primary outcome

The overlooked research gap in this domain is classified into three primary categories:

- Imperative need for a pipeline to preprocess narrative EMR for the identification of HAE: Current preprocessing tools such as NLTK [83], SpaCy [84], Stanza [85], and NegEx [86] are primarily designed for general text processing and do not specifically cater to medical text. Although there are specialized tools for

[1] Chunking involves identifying sequences of tokens based on parts of speech, employing rule-based models [79].

Table 3
Feature engineering approach, tools, pipelines, and medical lexicon.

Ref.	IE[a]	Vectorization[b]	Tools, pipelines and medical lexicon
[26]	FHS, NEG_E	–	MedLee
[28]	KS, NEG_B	–	–
[29]	KS		–
[30]	FHS, KS, NEG_B	–	Weighted regular expressions, CHARTextract
[31]	CH, FHS, IWR, KS, NER, NEG_B, NG, SWR	–	EasyCIE-PEDVT
[32]	CH, FHS, KS, NER, NEG_B, NG, SWR	–	IDEAL-X
[33]	–	BOW	–
[34]	NER, NEG_B	–	SNOMED-CT
[35]	KS	–	SNOMED-CT
[36]	KS, NEG_B	–	NegEx, UMLS, KMCI
[37]	KS	–	SNOMED-CT
[38]	–	–	–
[39]	–	–	–
[40]	KS	–	SNOMED-CT
[41]	KS, FHS, NER, NEG_B	–	UMLS
[42]	KS (chunking), prompting, FHS, NG	–	Gemma, Llama3, Mistral, Phi3
[43]	KS, NEG_B	–	–
[44]	NG	TF–IDF	NLTK and Scikit-learn
[45]	KS, prompting	–	GPT-4
[46]	KS, prompting	–	GPT-3.5
[47]	–	TF–IDF	–
[48]	SWR	CountVectorizer	–
[49]	COT, prompting	–	GPT-4.0, Qwen2, and Llama3.1
[50]	Prompting	–	GPT
[51]	NER	–	GPT-3
[52]	CH, FHS, NER, NEG_B, NG, SWR	–	CTAKES
[53]	KS, NEG_B, SWR	–	–
[54]	IWR, NEG_B	–	–
[55]	KS, NEG_B	–	PyConText
[56]	SWR	BOW, TF–IDF	–
[57]	CH, NER, NEG_B, NG, SWR	–	UMLS, CTAKES
[58]	–	BOW, BC, TF, TF–IDF	UMLS, MetaMap
[59]	Prompting	–	GPT-4
[60]	KS	–	–
[61]	KS, NER, NEG_B	–	UMLS, MetaMap
[62]	FHS, KS, NER	TF–IDF, BWC	–
[63]	BWC, FHS, KS, NEG_E, SWR	TF–IDF, WE	–
[64]	Chunking, prompting	–	Mixtral
[65]	COT, prompting	–	GLM-4.0, ERNIE Bot-3.5, Qwen-2.5, and GPT-3.5
[66]	KS, SWR	TF–IDF	–
[67]	NER	–	SpaCy
[68]	NG	BOW	RegEx, SNOMED-CT, Scikit-learn
[69]	CH, FHS, IWR, SWR	TF–IDF, W2V	–
[70]	NEG_B, NG	–	–
[71]	KS, NER, NEG_B	–	ScispaCy, UMLS, NegEx
[72]	–	BOW with N-gram, UWE	–
[73]	CH, IWR, KS, NER, NEG_B, SWR, NG	–	UMLS, MetaMap, LOD, SentiWordNet, LIBSVM
[74]	–	BOW	–

[a] "SWR" = "Stop words removal", "CH" = "Chunking with part-of-speech tagging", "NER" = "Named entity recognition", "IWR" = "Infrequent words removing", "FHS" = "Filtering header section", "KS" = "Keywords search", "NG" = "Extracting N-grams based on shallow parsing", "NEG_E" = "Explicit negation", "NEG_B" = "Explicit and implicit negations", "COT" = "Chain-of-Thought".
[b] "BOW" = "Bag-of-Words", "W2V" = "Word2Vec", "TF-IDF" = "Term frequency-inverse document frequency", "TF" = "Term frequency", "BC" = "Binary count", "BWC" = "BioWordVec", "UWE" = "Unigram word embedding", "WE" = "Word embedding".

preprocessing of narrative texts in EMR, including CTAKES [87], MedspaCy [88], BioStanza [89], ScispaCy [90], and RadText [91], these tools are not explicitly tailored for HAE detection to serve as a pipeline for detecting HAE.

To date, four pipelines have been developed that preprocess textual data to detect HAE, namely MedLee [80], PyConText [81], IDEAL-X [32], and EasyCIE-PEDVT [31]. MedLEE [80] is a rule-based NLP system for extracting structured medical concepts from clinical narratives, with capabilities for section and concept detection, basic medical term normalization, and limited temporal relation handling. PyConText is a Python-based NLP pipeline that enhances the capabilities of the ConText algorithm for processing clinical text. Chapman et al. employed PyConText to classify radiology reports for the identification of PE [81]. This system is rule-based, and its rules are designed to effectively capture the contextual nuances of findings throughout entire reports, as radiology reports frequently present multiple, sometimes conflicting, pieces of evidence regarding the presence of PE. This pipeline specifically addresses attributes such as negation (e.g., "no evidence of disease"), temporality (e.g., "prior condition"), and uncertainty (e.g., "possible infection").

In contrast, IDEAL-X [32] and EasyCIE-PEDVT [31] are rule-based pipelines developed for preprocessing narrative EMR data to detect VTE, including DVT and PE. Although both systems have demonstrated superior performance in identifying VTE, they are constrained by their rule-based nature, which results in an elevated incidence of false positives and limits the extraction of static concepts such as gender, disease, therapy, and medication. The increased false positive rate in rule-based systems stems from their reliance on rigid, predefined rules, which often lack the flexibility necessary to adapt to nuanced clinical cases [7,92,93]. Furthermore, rule-based approaches can misinterpret context, erroneously classifying past medical mentions as active conditions, thereby producing additional errors. Consequently, there remains a pressing need for a dedicated pipeline

Table 4
Major challenges in preprocessing narrative EMR.

Challenges	Limitations in HAE detection due to the nature of narrative texts in EMR data
Handling large volume of narrative EMR	• Narrative EMR reports length: The length of narrative EMR reports frequently exceeds the input limitations of LLMs models such as BERT, which can process a maximum of 512 words at any given time. This limitation presents challenges in capturing entire reports within a single input, potentially resulting in the omission or fragmentation of critical information necessary for detecting HAE (e.g., [74]). In the absence of effective section detection, the extraction and focus on relevant portions of the reports become problematic, thereby adversely affecting the model's capability to process the data in a comprehensive manner. • Absence of data integrity across sources: The absence of a standardized method for section detection within various types of textual notes results in inefficiencies when integrating data from multiple EMR sources. This deficiency in data integrity can lead to incomplete or fragmented information being entered into predictive models, thereby diminishing the accuracy and reliability of predictions. Such fragmentation hampers the ability to provide a coherent overview of patient information across different note types (e.g., [31,33]).
Text normalization	• Handling missing data: Inconsistent management of missing data across patient records adversely affects the accuracy of HAE detection (e.g., [34,52,74]). When missing data is not addressed uniformly, predictive models may encounter difficulties in interpreting incomplete records, potentially leading to biases and inaccuracies in HAE detection. • Absence of normalized clinical text: The diversity in writing styles and significant variability in the structure and vocabulary of narrative EMR data complicate the standardization of information. This absence of normalization hinders the capability of models to interpret and process data effectively, as inconsistencies in language and terminology can impede the model's ability to identify patterns critical for accurate HAE detection (e.g., [28,36,56,60,67,72]). • Tokenization limitations: Inconsistencies in the tokenization of narrative EMR data introduce an additional challenge. Effective tokenization is essential for segmenting text into manageable tokens for analysis. When tokenization techniques, such as unigrams or N-grams, are not uniformly applied, it may result in the fragmentation of meaningful phrases and a loss of contextual information (e.g., [33,53,55,82]). Inconsistent tokenization may impede the model's ability to comprehend relationships between words or phrases, thereby complicating the extraction of pertinent data for HAE detection.
Application of feature engineering	• Absence of temporal information: It is essential to extract temporal information from narrative EMR data in order to differentiate between past events and current occurrences, which is critical for the accurate detection of HAE. Without adequately capturing the temporal context, models may misinterpret the timing of events occurring during hospitalization or events that transpired prior to hospitalization. This oversight can lead to inaccuracies in the identification of HAE. Numerous studies have identified this limitation as a significant area necessitating improvement in preprocessing methods (e.g., [26,34,52]). • Vectorization limitations: The conversion of narrative EMR data from tokens into vector representations, such as embedding vectors, may result in a loss of information [74], particularly concerning laboratory data. The loss of a portion of information adversely impacts the accuracy of models, as critical nuances in clinical measurements and treatment regimens may not be adequately captured. Furthermore, a reliance solely on NLP techniques for IE may introduce biases or inherent limitations (e.g., [29,67]), as NLP models frequently falter in the interpretation of numerical or structured clinical data without supplementary preprocessing.

for preprocessing narrative EMR to detect various categories of HAE.

Table 5 illustrates the types of algorithms employed by current tools and pipelines for various preprocessing tasks, including section detection, text normalization, IE, and the extraction of temporal events. These algorithms are categorized as rule-based, ML, or a combination of both, referred to as a hybrid approach. This classification highlights the methodological diversity of various tools and pipelines and illustrates how they employ distinct techniques to efficiently process and analyze clinical text data. Furthermore, Table 5 indicates that the available tools are primarily designed for general tasks and lack customization for the detection of HAE (e.g. NLTK [83], SpaCy [84], Stanza [85], NegEx [86], MedLee [80], CTAKES [87], MedspaCy [88], BioStanza [89], ScispaCy [90], and RadText [91]). The limited number of pipelines specifically tailored for HAE detection typically address only a particular variant of HAE, including MedLee [80], PyConText [81], IDEAL-X [32], and EasyCIE-PEDVT [31]; all of these pipelines employ rule-based systems. However, the constraints of these rule-based systems encompass difficulties in adapting to the diverse and intricate nature of clinical language, challenges in managing ambiguities, and the requirement for extensive manual creation and maintenance of rules to ensure their continued effectiveness. The existing four pipelines exhibit a lack of uniformity in their architecture, as they are specifically designed for a particular type of HAE. This shortcoming highlights the pressing need for a pipeline to preprocess narrative texts in EMR, facilitating the identification of HAE. This requirement is identified as a research gap in Section 4.2.1.

The temporality feature is crucial for detecting HAE. Therefore, it is essential to consider the temporality aspect of each piece of extracted information. This feature indicates whether the information pertains to events that occurred prior to the patient's hospital visit or to adverse events that transpired during hospitalization. Consequently, the temporality of each tool and pipeline is also addressed in Table 5.

- Necessity for a robust system capable of managing the extensive volume of narrative EMR data: The proliferation of multiple data resources for narrative EMR data has resulted in an increasingly large, complex, and heterogeneous landscape, posing significant challenges in utilizing this data without effective management of its volume and customization for specific applications. Consequently, there is a critical need for a system that first addresses the challenges of integrating unstructured data [94,95] and subsequently manages the substantial volume of data to tailor it for specific purposes, such as HAE.
  While clinical tools such as CTAKES [87] and MedspaCy [88] have made commendable advances in section detection within clinical texts and attempt to manage the large volumes of data through section filtering, there remains a pressing requirement to effectively process this data to select information relevant to specific objective like HAE detection.
- Requirement for temporal event system for HAE detection: The identification of HAE is contingent upon the detection of incidents that transpire during hospitalization, while excluding any pre-existing patient conditions. Differentiating between adverse events that stem from pre-existing conditions and those that occur during hospitalization or within 30 days of readmission

Table 5
Comparison of tools and pipelines for text segmentation, text normalization, IE, and temporality detection.

Tools and pipelines	Section/Sentence detection[a]	Text normalization	IE[b]	Temporality detection
NLTK [83]	Rule-based; regular expression based rules for sentence detection.	Rule-based; includes stemming and lemmatization.	Rule-based; limited to basic POS tagging, NER.	Temporality detection not available.
SpaCy [84]	Hybrid; rule-based segmentation with dependency parsing.	Hybrid; rule-based with ML-based for lemmatization.	Hybrid; uses rule-based and ML for POS tagging, NER, dependency parsing.	Temporality detection not available.
Stanza [85]	ML-based (supervised); neural network models for sentence segmentation.	ML-based; applies neural models for lemmatization.	ML-based (supervised); for POS and NER.	Temporality detection not available
NegEx [86]	Rule-based; focused on negation detection, not full segmentation.	None specific for text normalization.	Rule-based; negation detection for clinical terms	Temporality detection not available.
MedLee [80]	Rule-based; focuses on section and concept detection.	Rule-based; basic normalization for medical terms.	Rule-based; designed for medical concept extraction.	Rule-based for limited temporality detection.
CTAKES [87]	Rule-based; rule-based segmentation with UIMA framework	Rule-based; uses dictionary-based normalization.	Hybrid; rule-based and some ML for clinical concept extraction.	Temporality detection not available.
MedspaCy [88]	Hybrid; rule-based (for section detection) and ML-based (for sentence detection).	Hybrid; custom rules and ML for normalization.	Hybrid; rule-based for clinical terms with ML extensions.	Temporality detection not available.
BioStanza [89]	ML-based (supervised); adapted from Stanza for sentence detection.	ML-based; applies neural models for lemmatization.	ML-based (supervised); for biomedical POS and NER.	Temporality detection not available.
ScispaCy [90]	Rule-based; rule-based algorithm with enhancements from dependency parsing, for sentence detection.	Hybrid; combines rule-based and ML techniques.	Hybrid; rule-based and ML models for scientific terms.	No temporality detection for extracted information.
RadText [91]	Hybrid; Rule-based (for section detection) and ML-based (for sentence detection).	ML-based; trained on medical text normalization.	Hybrid; uses ML for concept extraction with rule-based support.	Temporality detection not included.
IDEAL-X [32]	Rule-based; rule-based (for section detection and sentence detection).	Rule-based; rule-based for normalization.	Rule-based approach for specific clinical tasks.	Rule-based for temporality.
EasyCIE-PEDVT [31]	Hybrid; Rule-based (for section detection) and hybrid (for sentence detection).	Rule-based for normalization.	Hybrid; ML for entity recognition with rule-based context checks.	Rule-based for temporality.
PyConText [81]	Rule-based; context detection in sentences (not full segmentation).	None specific for text normalization.	Rule-based; focuses on sentence-level context detection.	Rule-based for limited temporality.

[a] "UIMA" = "Unstructured Information Management Architecture", "SS" = "Sentence segmentation".
[b] "NER" = "Named entity recognition", "POS" = "Part-Of-Speech tagging".

due to medical errors in narrative EMR data poses a considerable challenge. The aforementioned 30-day timeframe is established by the Hospital Readmission Reduction Program (HRRP), initiated under the Affordable Care Act in 2012, which enforces financial penalties on hospitals that experience excessive rates of readmission [96,97].

For example, if a patient presents with VTE upon admission and has not been hospitalized in the previous 30 days, this adverse event would not be classified as a HAE. The etiology of this VTE may be associated with pre-existing conditions, such as cancer [98,99] or obesity [98]. In contrast, if a patient develops VTE during their hospital stay, this occurrence would indeed be categorized as an HAE. Furthermore, if a patient has been hospitalized within the past 30 days and has undergone surgical procedures, any VTE identified at the time of admission would also fall under the definition of an HAE. These nuances underscore the challenges involved in distinguishing whether an adverse event qualifies as a true HAE or is merely a consequence of pre-existing health conditions.

One of the principal challenges associated with identifying temporal events from narrative EMR data is the prevalence of grammatical errors within these text [7]. Such data often encompass phrases with grammatical errors, incorrect verb tenses, and misapplied temporal adverbs, which complicate the establishment of timelines for the events experienced by patients. Furthermore, obstacles arising from limitations in data linkage across diverse healthcare settings hinder the ability to monitor a patient's condition over various time periods. Additionally, the extraction of comorbidities and the detection of present symptoms related to HAE, which frequently overlap with complex underlying patient conditions, further complicate this process [29,43,71]. This scenario highlights the urgent necessity of specifying temporal events in clinical texts to accurately identify HAE. Although a ML-based predictive model has been developed to project six-month VTE recurrence in cancer patients [34], the efficacy of this model remains uncertain. To date, EasyCIE-PEDVT [31] stands as the only preprocessing pipeline that has effectively incorporated temporal events for the detection of HAE.

The review delineates significant deficiencies in the preprocessing of narrative EMR data for the identification of HAE utilizing NLP and ML algorithms. Critical areas of concern include the necessity for a comprehensive preprocessing pipeline, the establishment of a system capable of managing substantial volumes of narrative EMR data, and the implementation of efficient temporal event detection mechanisms to facilitate HAE identification. Addressing these deficiencies is imperative for enhancing the accuracy and efficiency of HAE detection within clinical environments.

4.2.2. Secondary outcome

The secondary outcome focuses on synthesizing responses to research questions and evaluating the efficacy of HAE detection influenced by preprocessing tasks. The evaluation of HAE detection using

Table 6
Evaluating the performance of HAE detection based on the preprocessing techniques.

Ref.[a]	Eval.[b]	HAE types[c]	Data source[d]	Setting[e]	HAE detection approach[f]	Normalization Tech.[g]	IE Tech.[h]	Vectorization[i]	Evaluation[i]
[55]	Ev.A01	Intra-abd SSI	Public (MIMIC-III)/665 RR	BIDMC	ICD-9 codes	–	–	–	F1 (48%)
	Ev.A02	Intra-abd SSI	Public (MIMIC-III)/665 RR	BIDMC	SVM	N-grams	–	–	F1 (76%)
	Ev.A03	Intra-abd SSI	Public (MIMIC-III)/665 RR	BIDMC	fcFinder	–	–	–	F1 (66%)
	Ev.A04	Intra-abd SSI	Public (MIMIC-III)/665 RR	BIDMC	fcFinder	–	NEG_B	–	F1 (79%)
	Ev.A05	Intra-abd SSI	Public (MIMIC-III)/665 RR	BIDMC	fcFinder	–	NEG_B,KS	–	F1 (88%)
[33]	Ev.B01	DVT	Private/2000 RR	MUHC	NLP+SVM (linear kernel)	SeChr and CF	–	BOW	AUC-ROC (95%)
	Ev.B02	DVT	Private/2000 RR	MUHC	NLP+SVM (linear kernel)	SeChr, CF, unigram and bigrams	–	BOW	AUC-ROC (96%)
	Ev.B03	DVT	Private/2000 RR	MUHC	NLP+SVM (linear kernel)	SeChr, CF, N-grams, and Lemma	–	BOW	AUC-ROC (98%)
	Ev.B04	DVT	Private/2000 RR	MUHC	NLP+SVM (RBF kernel)	SeChr and CF	–	BOW	AUC-ROC (95%)
	Ev.B05	DVT	Private/2000 RR	MUHC	NLP+SVM (RBF kernel)	SeChr, CF, unigram and bigrams	–	BOW	AUC-ROC (97%)
	Ev.B06	DVT	Private/2000 RR	MUHC	NLP+SVM (RBF kernel)	SeChr, CF, N-grams, and Lemma	–	BOW	AUC-ROC (98%)
	Ev.B07	PE	Private/2000 RR	MUHC	NLP+SVM (linear kernel)	SeChr and CF	–	BOW	AUC-ROC (92%)
	Ev.B08	PE	Private/2000 RR	MUHC	NLP+SVM (linear kernel)	SeChr, CF, unigram and bigrams	–	BOW	AUC-ROC (95%)
	Ev.B09	PE	Private/2000 RR	MUHC	NLP+SVM (linear kernel)	SeChr, CF, N-grams, and Lemma	–	BOW	AUC-ROC (99%)
	Ev.B10	PE	Private/2000 RR	MUHC	NLP+SVM (RBF kernel)	SeChr and CF	–	BOW	AUC-ROC (95%)
	Ev.B11	PE	Private/2000 RR	MUHC	NLP+SVM (RBF kernel)	SeChr, CF, unigram and bigrams	–	BOW	AUC-ROC (96%)
	Ev.B12	PE	Private/2000 RR	MUHC	NLP+SVM (RBF kernel)	SeChr, CF, N-grams, and Lemma	–	BOW	AUC-ROC (99%)
[53]	Ev.C01	SSI	Private/1407 patients	SMHNY	LR (Variables from administrative data)	–	SWR, NEG_B	–	PPV (70%), Sen. (68%) and F1 (66%)
	Ev.C02	SSI	Private/1407 patients	SMHNY	LR	–	KS, SWR, NEG_B	–	PPV (93%), Sen. (93%) and F1 (93%)
	Ev.C03	SSI	Private/1407 patients	SMHNY	LR	Lemma	KS, SWR, NEG_B	–	PPV (95%), Sen. (95%) and F1 (95%)
	Ev.C04	SSI	Private/1407 patients	SMHNY	LR	5-g	SWR, NEG_B	–	PPV (97%), Sen. (97%) and F1 (97%)
	Ev.C05	SSI	Private/1407 patients	SMHNY	LR (Variables from administrative data)	–	KS, SWR, NEG_B	–	PPV (92%), Sen. (92%) and F1 (92%)
	Ev.C06	SSI	Private/1407 patients	SMHNY	LR (Variables from administrative data)	Lemma	KS, SWR, NEG_B	–	PPV (96%), Sen. (96%) and F1 (96%)
	Ev.C07	SSI	Private/1407 patients	SMHNY	LR (Variables from administrative data)	5-g	SWR, NEG_B	–	PPV (97%), Sen. (97%) and F1 (97%)
[58]	Ev.D01	Medications	Private (Benchmark_AIMS)/564 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (67.4%)
	Ev.D02	Medications	Private (Benchmark_AIMS)/564 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (78.5%)
	Ev.D03	Medications	Private (Original_AIMS)/142 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (76.3%)
	Ev.D04	Medications	Private (Original_AIMS)/142 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (87.1%)
	Ev.D05	Medications	Private (Independent_Riskman)/2270 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (75.2%)
	Ev.D06	Medications	Private (Independent_Riskman)/2270 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (85.1%)
	Ev.D07	Falls	Private (Benchmark_AIMS)/521 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (88.1%)
	Ev.D08	Falls	Private (Benchmark_AIMS)/521 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (88.1%)
	Ev.D09	Falls	Private (Original_AIMS)/181 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (89.7%)
	Ev.D10	Falls	Private (Original_AIMS)/181 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (92.4%)
	Ev.D11	Falls	Private (Independent_Riskman)/1811 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (81%)
	Ev.D12	Falls	Private (Independent_Riskman)/1811 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (90.3%)
	Ev.D13	Pressure injury	Private (Benchmark_AIMS)/524 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (91.5%)
	Ev.D14	Pressure injury	Private (Benchmark_AIMS)/524 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (96.4%)
	Ev.D15	Pressure injury	Private (Original_AIMS)/75 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (85.4%)
	Ev.D16	Pressure injury	Private (Original_AIMS)/75 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (91.6%)
	Ev.D17	Pressure injury	Private (Independent_Riskman)/387 IR	Australia	SVM (RBF kernel)	SeChr, CF	–	BOW	F (81.4%)
	Ev.D18	Pressure injury	Private (Independent_Riskman)/387 IR	Australia	SVM (RBF kernel)	SeChr, CF	Semantic feature extraction	BOC	F (85.7%)
[73]	Ev.E01	VTE	Public (TREC-CDS and i2b2)/150 narrative reports	–	SVM (RBF kernel)	–	–	–	F1 (36.4%)
	Ev.E02	VTE	Public (TREC-CDS and i2b2)/150 narrative reports	–	SVM (RBF kernel)	SeChr, POS, N-gram, Synonyms	Semantic feature extraction	–	F1 (66.7%)
[31]	Ev.F01	DVT	Private/2266 textual notes in internal dataset	–	EasyCIE-PEDVT	SeChr, POS, N-gram, Synonyms	NER, SWR, CH, IWR, NG, FHS, KS, NEG_B	–	AUC-ROC (92%)
	Ev.F02	DVT	Private/2266 textual notes in internal dataset	–	ICD codes	–	–	–	AUC-ROC (76.1%)
	Ev.F03	DVT	Private/16,493 textual notes in external dataset	–	EasyCIE-PEDVT	SeChr, POS, N-gram, Synonyms	NER, SWR, CH, IWR, NG, FHS, KS, NEG_B	–	AUC-ROC (92.1%)
	Ev.F04	DVT	Private/16,493 textual notes in external dataset	–	ICD codes	–	–	–	AUC-ROC (79.4%)
	Ev.F05	PE	Private/2266 textual notes in internal dataset	–	EasyCIE-PEDVT	SeChr, POS, N-gram, Synonyms	NER, SWR, CH, IWR, NG, FHS, KS, NEG_B	–	AUC-ROC (85.4%)
	Ev.F06	PE	Private/2266 textual notes in internal dataset	–	ICD codes	–	–	–	AUC-ROC (84%)
	Ev.F07	PE	Private/16,493 textual notes in external dataset	–	EasyCIE-PEDVT	SeChr, POS, N-gram, Synonyms	NER, SWR, CH, IWR, NG, FHS, KS, NEG_B	–	AUC-ROC (90.7%)
	Ev.F08	PE	Private/16,493 textual notes in external dataset	–	ICD codes	–	–	–	AUC-ROC (95.1%)
[69]	Ev.G01	IPF	Private/665,156 textual notes from Stanford dataset	IRB	XGBoost	SeChr, POS, N-gram, CF	SWR, CH, IWR, FHS	TF–IDF	AUC-PR (66.8%)
	Ev.G02	IPF	Private/665,156 textual notes from Stanford dataset	IRB	XGBoost	SeChr, POS, N-gram, CF	SWR, CH, IWR, FHS	BioBERT	AUC-PR (76%)
	Ev.G03	IPF	Private/665,156 textual notes from Stanford dataset	IRB	XGBoost	SeChr, POS, N-gram, CF	SWR, CH, IWR, FHS	W2V (skip-gram)	AUC-PR (84.2%)
	Ev.G04	IPF	Public (MIMIC-III)/91.7M notes	BIDMC	XGBoost	SeChr, POS, N-gram, CF	SWR, CH, IWR, FHS	TF–IDF	AUC-PR (64.2%)
	Ev.G05	IPF	Public (MIMIC-III)/91.7M notes	BIDMC	XGBoost	SeChr, POS, N-gram, CF	SWR, CH, IWR, FHS	BioBERT	AUC-PR (74.1%)
	Ev.G06	IPF	Public (MIMIC-III)/91.7M notes	BIDMC	XGBoost	SeChr, POS, N-gram, CF	SWR, CH, IWR, FHS	W2V (skip-gram)	AUC-PR (81.5%)

(*continued on next page*)

Table 6 (*continued*).

[56]	Ev.H01	RWI	Private/Free-text notes for 5860 patients	–	XGBoost	SeChr, CF, Lemma	SWR	BOW, TF–IDF	AUC-PR (96%)
	Ev.H02	RWI	Private/Free-text notes for 5860 patients	–	CPT and ICD codes	–	SWR	BOW, TF–IDF	AUC-PR (76%)
[38]	Ev.I01	VTE	Private/2273 RR	–	NLP+BERT	CF	–	–	AUC-PR (99%)
	Ev.I02	VTE	Private/2273 RR	–	NLP+BERT	CF, Synonyms	–	–	AUC-PR (99.5%)
	Ev.I03	VTE	Private/1358 RR (only Emory)	–	NLP+BERT	CF	–	–	AUC-PR (98.6%)
	Ev.I04	VTE	Private/1358 RR (only Emory)	–	NLP+BERT	CF, Synonyms	–	–	AUC-PR (98.8%)
	Ev.I05	VTE	Private/915 RR (only Grady)	–	NLP+BERT	CF	–	–	AUC-PR (99.4%)
	Ev.I06	VTE	Private/915 RR (only Grady)	–	NLP+BERT	CF, Synonyms	–	–	AUC-PR (99.4%)
[39]	Ev.J01	VTE/PE	Private/1900 RR	–	NLP+ClinicalBERT	–	–	–	Accuracy (89%)
	Ev.J02	VTE/PE	Private/1900 RR	–	NLP+ClinicalBERT	Synonyms (Random swapping)	–	–	Accuracy (90%)
	Ev.J03	VTE/PE	Private/1900 RR	–	NLP+ClinicalBERT	Synonyms (Replacement)	–	–	Accuracy (91%)
[42]	Ev.K01	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	ICD	–	–	–	F1 (20.93%)
	Ev.K02	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Gemma)	POS, N-gram	KS	–	F1 (28.11%)
	Ev.K03	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Gemma)	POS, N-gram	Semantic similarity	–	F1 (22.03%)
	Ev.K04	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Llama3)	POS, N-gram	KS	–	F1 (24.65%)
	Ev.K05	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Llama3)	POS, N-gram	Semantic similarity	–	F1 (20.31%)
	Ev.K06	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Mistral)	POS, N-gram	KS	–	F1 (24.62%)
	Ev.K07	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Mistral)	POS, N-gram	Semantic similarity	–	F1 (20.83%)
	Ev.K08	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Phi3)	POS, N-gram	KS	–	F1 (23.39%)
	Ev.K09	PE	Private/10,066 patients	Tertiary-care hospitals in Calgary, Canada	LLM (Phi3)	POS, N-gram	Semantic similarity	–	F1 (17.06%)
[64]	Ev.L01	Sepsis	Private/754 patients (Cohort B)	UCSD health hospital	LLM (Mixtral)	–	Likelihood tools[k]	–	F1 (44.2%)
	Ev.L02	Sepsis	Private/754 patients (Cohort B)	UCSD health hospital	LLM (Mixtral)	–	Chunking	–	F1 (61%)
	Ev.L03	Sepsis	Private/1746 patients (Cohort A)	UCSD health hospital	LLM (Mixtral)	–	Likelihood tools[k]	–	F1 (47.9%)
	Ev.L04	Sepsis	Private/1746 patients (Cohort A)	UCSD health hospital	LLM (Mixtral)	–	Chunking	–	F1 (63.9%)

[a] "Ref." = "Reference".

[b] "Eval." = "Evaluation number".

[c] "ADR" = "Adverse drug reaction", "Post." = "Postoperative", "PWD" = "Postoperative wound dehiscence", "UTI" = "Urinary tract infection", "Intra-abd" = "Intra-abdominal", "SSI" = "Surgical site infections", "IPF" = "Insulin pump failure", "RWI" = "Reoperation for wound infection".

[d] "RR" = "Radiology report", "IR" = "Incident reports".

[e] "4MCRLI" = "4 medical centers in the Region of Lombardy in Italy", "BIDMC" = "Beth Israel Deaconess Medical Center in Boston", "MUHC" = "McGill University Health Center", "SMHNY" = Strong Memorial Hospital in New York, "IRB" = Stanford Institutional Review Board, "TREC-CDS" = Text Retrieval Conference Clinical Decision Support.

[f] "KS" = "Keyword search", "ICD" = "International Classification of Diseases", "SVM" = "Support Vector Machine", "LR" = "Logistic Regression", "CPT" = "Current procedural terminology".

[g] "Tech." = "technique", "SeChr" = "Special characters", "CF" = "Case folding", "Abbr" = "Abbreviation", "Lemma" = "Lemmatization and/or stemming".

[h] "Tech." = "technique", "SWR" = "Stop words removal", "CH" = "Chunking and part-of-speech tagging", "IWR" = "Infrequent words removing", "FHS" = "Filtering header section", "KS" = "Keywords search", "NG" = "Extracting N-grams based on shallow parsing", "NEG_E" = "Explicit negation", "NEG_B" = "Explicit and implicit negations".

[i] "BOW" = "Bag-of-Words", "BOC" = "Bag-of-Concepts", "TF-IDF" = "Term frequency-inverse document frequency", "W2V" = "Word2Vec".

[j] "PPV" = "Positive predictive value", "F" = "F-score", "F1" = "F1-score", "Sen." = "Sensitivity", "AUC-ROC" = "Area Under the Receiver Operating Characteristic Curve", "AUC-PR" = "Area Under the Curve Precision-Recall".

[k] The study utilized sepsis likelihood tools to extract the relevant information.

narrative EMR is divided into three groups: (1) studies using only text normalization, (2) studies focusing solely on feature engineering, and (3) studies incorporating both text normalization and feature engineering in preprocessing. The 48 studies demonstrated that applying an NLP approach enhances the performance of HAE detection. Among the 48 studies analyzed, none examined the influence of data volume on the detection of HAE. Moreover, only 12 studies, including [31,33,38,39,42,53,55,56,58,64,69,73], integrated various preprocessing methodologies to enhance detection performance through the application of different NLP techniques to narrative EMR. In contrast, the remaining 36 studies employed NLP as a preprocessing step for HAE detection but did not assess the performance of HAE detection with and without the application of specific NLP techniques. Collectively, these 12 studies conducted a total of 114 experiments.

It is clear that factors beyond preprocessing have an impact on the performance of HAE detection. These factors encompass the diversity of HAE types, such as VTE (including DVT and PE) and postoperative sepsis, as well as variations in data sources, institutional settings, and specific technical approaches to HAE detection. These approaches may include keyword searches, ML models, or the use of International Classification of Diseases (ICD) codes. Given these variations, comparing HAE detection performance across studies that operate within different institutional contexts, utilize diverse data sources, and focus on varying HAE types can lead to inconsistent and potentially misleading outcomes. Therefore, a meaningful comparison necessitates the control of variables including HAE type, data source, institutional setting, and HAE detection approach. This allows preprocessing techniques to be examined as the independent variable, while HAE detection performance serves as the dependent variable. Such a methodology facilitates a standardized and controlled comparison across studies, thereby ensuring a clearer and more accurate evaluation of the influence of preprocessing techniques.

Table 6 presents an assessment of HAE detection performance, organized by various preprocessing techniques. To facilitate equitable comparisons under consistent conditions, the second column delineates a unique evaluation number for each experiment. This numbering system allows for a clear discussion and comparison of experiments conducted under the same conditions but utilizing different preprocessing techniques.

- **Text normalization**: Research indicates that the normalization of text significantly improves the performance of HAE detection. A comparison of the experiments conducted by Chapman et al. illustrates notable enhancements in accuracy when employing text normalization preprocessing, in contrast to its absence [55]. For instance, in the detection of intra-abdominal SSI using only ICD-9 codes, the F1-score was restricted to 48%, thereby underscoring the inherent limitations of this fundamental approach. Conversely, when utilizing a Support Vector Machine (SVM) algorithm for HAE detection with N-grams normalization, the F1-score markedly increased to 76% under controlled conditions, as evidenced by the comparison between evaluation numbers "Ev.A01" and "Ev.A02" in Table 6.

  Some experiments have been conducted using a rule-based NLP pipeline, fcFinder, which is based on PyConText for extracting clinical information relevant to the detection of SSI. The utilization of fcFinder, absent any preprocessing techniques and relying

exclusively on keyword searches, resulted in an increase in the F1-score to 66%. This improvement is significant when compared to SSI detection employing ICD-9 codes under equivalent conditions, as evidenced by the evaluation numbers "Ev.A01" and "Ev.A03" in Table 6. This development highlights the critical role of keyword searches within algorithms in enhancing the accuracy of HAE detection.

A study conducted by Rochefort et al. [33] underscores the significance of text normalization as a preprocessing step to enhance the performance of HAE detection. In the context of DVT detection, the analysis detailed in Table 6, specifically the comparison between evaluations "Ev.B01" and "Ev.B02", illustrates that the incorporation of unigram and bigram tokenization techniques in preprocessing contributes to improved model performance. This is evidenced by an increase in the Area Under the Receiver Operating Characteristic (AUC-ROC) from 95% to 96%. These findings suggest that incorporating basic N-gram features into text processing has a positive impact on DVT detection outcomes.

Further enhancements are noted when contrasting "Ev.B01" with "Ev.B03" in Table 6. The application of advanced text normalization techniques, which include both N-grams and lemmatization, results in an AUC-ROC of 98%, signifying a notable improvement in model accuracy. Additionally, the analysis of "Ev.B02" and "Ev.B03" reinforces the effectiveness of these advanced normalization techniques, demonstrating a significant increase in DVT detection performance, reflected by an AUC-ROC rise from 96% to 98%. These findings emphasize the crucial role of robust text normalization in preprocessing EMR to accurately identify DVT as a type of HAE.

Furthermore, in regard to PE detection, the implementation of unigram and bigram features in "Ev.B08", relative to "Ev.B07" in Table 6, elevates the AUC-ROC from 92% to 95%, thus highlighting the advantages of employing simple N-grams features in preprocessing. The transition from "Ev.B07" to "Ev.B09" demonstrates an additional increase in the AUC-ROC to 99%, indicating a substantial enhancement in detection performance due to the combination of N-grams and lemmatization. The comparison between "Ev.B08" and "Ev.B09" further corroborates these observations, as the inclusion of additional normalization techniques elevates the AUC-ROC from 95% to 99%. Collectively, these results underscore the significant contribution of text normalization techniques, particularly those involving N-grams and lemmatization, in advancing clinical model accuracy for PE detection.

Thirukumaran et al. conducted a series of experiments utilizing 5-g, stemming, and lemmatization for tokenization and normalization and keyword search approach for IE, as documented in their research [53]. In a controlled environment, the comparison of "Ev.C01" (employing the Logistic Regression (LR) algorithm based solely on administrative data) against "Ev.C04" (which integrates 5-g) demonstrated a significant enhancement in the F1-score for the detection of SSI. Specifically, "Ev.C01" yielded an F1-score of 66%, whereas "Ev.C04" achieved a substantially higher F1-score of 97%. This disparity highlights the significant impact of incorporating text normalization techniques on improving detection performance.

Further analyses reinforce these findings: the implementation of additional text normalization methods, such as lemmatization (Ev.C03), 5-g (Ev.C04), keyword search combined with administrative data (Ev.C05), and the integration of both keyword search and lemmatization (Ev.C06), consistently resulted in progressively improved F1-scores. These results illustrate that normalization approaches significantly enhance SSI detection, as evidenced by the comparisons made between "Ev.C01" and "Ev.C03", "Ev.C01" and "Ev.C04", "Ev.C01" and "Ev.C05", "Ev.C01" and "Ev.C06", as well as "Ev.C01" and "Ev.C07" in Table 6. The examination of "Ev.C02" in comparison to "Ev.C03" reveals an additional increase in the F1-score upon the inclusion of lemmatization, thereby suggesting that this technique contributes positively to detection performance.

Moreover, the comparison between "Ev.C02" and "Ev.C04" illustrates a boost in the F1-score with the application of 5-g, indicating its strong impact on performance. A further examination contrasting "Ev.C03" with "Ev.C04" substantiates that the use of 5-g surpasses the efficacy of lemmatization when considered in isolation. Additional corroboration is provided in the comparisons of "Ev.C05" with "Ev.C07" and "Ev.C06" with "Ev.C07", highlighting that the combination of lemmatization and 5-g yields optimal performance enhancements in SSI detection. It is noteworthy that, while the findings suggest that N-grams tend to outperform lemmatization in these experiments, this conclusion is drawn from a limited scope of seven experiments. Consequently, further research with a broader range of experimental designs would be essential to conclusively ascertain whether N-grams consistently deliver superior performance relative to lemmatization, although it is evident that both techniques significantly improve SSI detection. Certain studies have utilized synonym replacement techniques within the text to normalize the data and enhance augmentation prior to its input into ML algorithms [38,39]. As presented in Table 6, the comparison between ("Ev.I01" and "Ev.I02") and ("Ev.I03" and "Ev.I04") in the study by Wang et al. [38], as well as the comparison between ("Ev.J01" and "Ev.J02") and ("Ev.J01" and "Ev.J03") in the research conducted by Deng et al. [39], demonstrates a significant improvement in the performance of HAE detection with the application of synonyms.

- **Feature engineering**: Feature engineering includes IE approaches as well as vectorization models. The findings indicate that studies utilizing IE techniques have shown enhancements in the performance of HAE detection, as evidenced by the works of [48,53,55,58,63,69,73]. For instance, the study conducted by Thirukumaran et al. demonstrated that in a controlled environment keyword search approach enhance the performance of HAE detection [53]. As shown in Table 6, the comparison of "Ev.C01" (employing the Logistic Regression (LR) algorithm based solely on administrative data) against "Ev.C05" (which integrates keyword search approach) demonstrated a significant enhancement in the F1-score for the detection of SSI. Specifically, "Ev.C01" yielded an F1-score of 66%, whereas "Ev.C05" achieved a substantially higher F1-score of 92%. This disparity underscores the considerable influence of incorporating feature engineering techniques on the improvement of HAE detection performance.

  Indeed, Wang et al. investigated ten distinct incident types, such as falls, pressure injuries, medication errors, documentation issues, and clinical handovers [58]. Their research revealed that the application of semantic relationships between concepts, facilitated by the Unified Medical Language System (UMLS) and MetaMap for semantic feature representation, significantly enhanced the performance of HAE detection. Across their sixty experiments, the integration of semantic IE consistently improved performance.

  In Table 6, only the results pertaining to falls, pressure injuries, and medication errors are presented for brevity, encompassing eighteen experiments. For example, the detection of adverse medication reactions in a controlled setting utilizing the same dataset (i.e., Benchmark dataset) yielded an F-score of 67.5% without semantic extraction, which increased to 78.5% with its application (referencing the evaluation numbers "Ev.D01" and "Ev.D02" in Table 6). Similarly, the evaluation conducted on an alternative dataset (i.e., original_dataset) showed an increase in F-score from 76.3% to 87.1% when semantic extraction was employed (comparing evaluation numbers "Ev.D03" and "Ev.D04" in Table

6). Additional analyses, including comparisons between "Ev.D07" and "Ev.D08", "Ev.D09" and "Ev.D10", as well as "Ev.D11" and "Ev.D12", all related to falls detection as HAE under consistent conditions in Table 6, further corroborate these results, evidencing superior performance with semantic extraction. Overall, the sixty experiments conducted in this study, which encompass thirty pairwise comparisons, underscore the beneficial impact of semantic extraction techniques on HAE detection [58].

An additional piece of evidence concerning the integration of semantic extraction and sentiment assessment techniques within IE during feature engineering highlights their efficacy in enhancing the performance of HAE detection, as presented in the study by Sabra et al. [73]. This research indicates that the application of these techniques in conjunction with SVM resulted in improved performance, evidenced by an F1-score of 66.7%. In contrast, the F1-score observed when utilizing SVM independently was 36.4% [73]. This enhancement is further detailed in the comparison of evaluation metrics "Ev.E01" and "Ev.E02" found in Table 6.

Moreover, the research conducted by Gupta et al. underscores the critical role of feature engineering in the identification of Immune-Related Adverse Events (irAEs), which represent a specific category of HAE [63]. In their investigation, the authors implemented ML algorithms to detect four classifications of irAEs, executing a total of 44 experiments. Their results illustrated that the combination of keyword filtering, an IE technique, with vectorization substantially improved the efficacy of irAE detection.

Sanyal et al. conducted a study comprising six experiments to assess the influence of vectorization methods, as part of feature engineering, on the detection of insulin pump failures, categorized as a type of HAE [69]. The study demonstrated that insulin pump failure detection in a controlled environment, utilizing the Stanford dataset, achieved an Area Under the Curve for Precision–Recall (AUC-PR) of 66.8% with the Term Frequency–Inverse Document Frequency (TF–IDF) representation method. This performance improved to 76% with the application of BioBERT, as evidenced by the comparative evaluation numbers "Ev.G01" and "Ev.G02" presented in Table 6. Further enhancement was observed, with performance reaching 84.2% using the Word-to-Vector (skip-gram) method, comparing evaluation numbers "Ev.G01" and "Ev.G03" in the same table. The comparisons between evaluation numbers "Ev.G04" and "Ev.G05", as well as "Ev.G04" and "Ev.G06", underscore the critical role of vectorization methods in improving HAE detection performance. While the Word-to-Vector (skip-gram) method exhibited the highest performance across these experiments (notably between "Ev.G02" and "Ev.G03", and "Ev.G05" and "Ev.G06"), reaching a definitive conclusion regarding its superiority as a vectorization method necessitates additional evidence and testing within controlled settings.

Further evidence supporting the significance of feature engineering in the detection of HAE is presented by Chapman et al. who employed the fcFinder NLP pipeline to integrate semantic and linguistic feature extraction techniques for the identification of intra-abdominal SSI [55]. By implementing negation within their IE methodologies, they attained an F1-score of 79% for SSI detection, marking a considerable enhancement compared to the 48% F1-score achieved through the use of ICD-9 codes under the same controlled conditions (as indicated by the comparative evaluation numbers "Ev.A04" and "Ev.A01" in Table 6). Moreover, the application of both negation and semantic feature extraction, utilizing keyword searches aligned with anatomical locations pertinent to SSIs, resulted in an F1-score that increased to 88% (referencing the comparison between evaluation numbers "Ev.A05" and "Ev.A01" in Table 6). This underscores the effectiveness of advanced NLP pipelines, in conjunction with semantic features, in enhancing detection accuracy.

- **Text normalization and feature engineering**: Shi et al. employed the EasyCIE-PEDVT NLP pipeline for predicting VTE, integrating NLP techniques for text normalization alongside feature engineering methods as part of their preprocessing approach [31]. Their investigation encompassed eight experiments, four of which indicated that the synergy of feature engineering and text normalization significantly improved detection performance for DVT and PE in comparison to traditional ICD codes. For instance, DVT detection within a controlled environment using a consistent dataset (i.e., internal dataset) achieved an area under the AUC-ROC of 76.1% when relying exclusively on ICD codes. Conversely, this AUC-ROC elevated to 92% when incorporating feature engineering and text normalization techniques via the EasyCIE-PEDVT pipeline, as illustrated by the comparison of evaluation numbers "Ev.F01" and "Ev.F02" in Table 6. Furthermore, additional evidence supporting the efficacy of combining text normalization with feature engineering in DVT detection is presented through the comparison of "Ev.F03" and "Ev.F04" in the same table. It is also noteworthy that no significant differences were observed between the discrimination performance of EasyCIE-PEDVT and ICD codes regarding PE detection, as evidenced by the comparisons of evaluation numbers "Ev.F05" and "Ev.F06", as well as between "Ev.F07" and "Ev.F08" within the same table.

  Further evidence supporting the combined application of NLP techniques for text normalization and feature engineering as preprocessing steps to enhance the detection of HAE is derived from a study conducted by Karhade et al. [56]. They developed a gradient-boosting NLP algorithm specifically designed to identify wound infections necessitating reoperation. This methodology achieved a AUC-PR of 96%, which markedly surpassed the detection results sourced from Current Procedural Terminology (CPT) and ICD codes without preprocessing, yielding a AUC-PR of merely 76% [56].

  Additionally, a study by Dante et al. employed a customized NLP pipeline to detect HAE, IDEAL-X, for text normalization and feature engineering aimed at improving the detection of VTE [32]. This methodology demonstrated a sensitivity of 97.6% and a specificity of 99.8% in detecting VTE.

A comprehensive analysis of methodologies for the detection of HAE utilizing narrative EMR data, supplemented by NLP and ML techniques, reveals that these approaches significantly outperform traditional HAE detection methods based on ICD codes or administrative data. The integration of narrative EMR and advanced preprocessing techniques notably enhances the performance of HAE detection. Comparative analyses reveal pronounced differences in performance across several scenarios: Evidence 1 [55] (comparison between "Ev.A01" and "Ev.A02"), Evidence 2 [55] (comparison between "Ev.A01" and "Ev.A03"), Evidence 3 [55] (comparison between "Ev.A01" and "Ev.A04"), Evidence 4 [55] (comparison between "Ev.A01" and "Ev.A05"), Evidence 5 [53](comparison between "Ev.C01" and "Ev.C02"), Evidence 6 [53] (comparison between "Ev.C01" and "Ev.C03"), and Evidence 7 [53] (comparison between "Ev.C01" and "Ev.C04"). These findings emphasize the critical role of preprocessing in augmenting the performance and reliability of HAE detection results.

Eight pieces of evidence indicate that the utilization of N-grams tokenization significantly enhances the efficacy of HAE detection. The following comparative analyses substantiate this assertion: Evidence 1, which involves a comparison between "Ev.B01" and "Ev.B02" [33]; Evidence 2, comparing "Ev.B04" and "Ev.B05" [33]; Evidence 3, which examines the difference between "Ev.B07" and "Ev.B08" [33]; Evidence 4, involving a comparison of "Ev.B10" and "Ev.B11" [33]; Evidence 5, the comparison between "Ev.C02" and "Ev.C04" [53]; Evidence 6, which assesses "Ev.C03" against "Ev.C04" [53]; Evidence 7, which compares "Ev.C05" and "Ev.C07" [53]; and finally, Evidence 8, involving the comparison of "Ev.C06" and "Ev.C07" [53].

In assessing lemmatization as an independent text normalization technique, the existing evidence regarding its efficacy in enhancing HAE detection performance is limited. Noteworthy findings include Evidence 1, which compares "Ev.C02" and "Ev.C03" [53]; Evidence 2, which examines the distinction between "Ev.C05" and "Ev.C06" [53]; and Evidence 3, which involves a comparison of "Ev.H01" and "Ev.H02" [56]. However, these results are insufficient to endorse lemmatization as a standalone approach for tokenization aimed at improving HAE detection performance.

Additional evidence indicates that the integration of N-grams tokenization and various normalization techniques, such as lemmatization or synonym identification, significantly enhances the performance of HAE detection. Relevant comparative analyses include: Evidence 1 (comparison between "Ev.B01" and "Ev.B03") [33], Evidence 2 (comparison between "Ev.B02" and "Ev.B03") [33], Evidence 3 (comparison between "Ev.B07" and "Ev.B09") [33], Evidence 4 (comparison between "Ev.B08" and "Ev.B09") [33], Evidence 5 (comparison between "Ev.B10" and "Ev.B12") [33], Evidence 6 (comparison between "Ev.B11" and "Ev.B12") [33], Evidence 7 (comparison between "Ev.F01" and "Ev.F02") [31], and Evidence 8 (comparison between "Ev.F03" and "Ev.F04") [31].

A total of 76 experiments indicate that semantic feature extraction significantly enhances the performance of HAE detection. Notably, 60 experiments (i.e. 30 evidences) referenced in [58] demonstrate improved outcomes when semantic feature extraction is employed during the preprocessing stage. Nine evidence are presented in Table 6, including comparisons such as "Ev.D07" versus "Ev.D08", "Ev.D09" versus "Ev.D10", and "Ev.D11" versus "Ev.D12". Furthermore, two additional pieces of evidence can be found in Chapman's study [55] by applying fcFinder: Evidence 1, which compares "Ev.A03" and "Ev.A05", and Evidence 2, which examines the comparison between "Ev.A04" and "Ev.A05". An additional piece of evidence is presented in Sabra's study [73], specifically identified as Evidence 3, through a comparative analysis of "Ev.E01" and "Ev.E02".

A study conducted by Cheligeer [42] presents six additional pieces of evidence regarding the application of LLMs in the detection of HAE. The findings illustrated in Table 6 underscore the significance of semantic feature extraction in identifying HAE, as exemplified by the comparisons between "Ev.K02" and "Ev.K01", as well as "Ev.K03" and "Ev.K01". Furthermore, the results in the Table 6 indicate that employing keyword, search as an IE technique yields superior performance compared to utilizing semantic similarity, particularly in instances such as "Ev.K02" versus "Ev.K03", "Ev.K04" versus "Ev.K05", "Ev.K06" versus "Ev.K07", and "Ev.K08" versus "Ev.K09".

A study conducted by Shashikumar [64] provides two additional pieces of evidence regarding the application of LLMs in the detection of HAE. The findings indicate that when relevant chunks pertaining to HAE are extracted from the clinical text, the performance of HAE detection is substantially improved. Comparative analyses between the pairs "Ev.L01" and "Ev.L02", as well as "Ev.L03" and "Ev.L04", are presented in Table 6.

The analysis of preprocessing narrative EMR data for the identification of HAE indicates that the application of N-grams tokenization, coupled with integrated methodologies such as normalizing N-grams through lemmatization or the use of synonyms, significantly enhances performance of HAE detection. Additionally, the implementation of semantic feature extraction as a preliminary processing step improves detection capabilities by effectively capturing deeper contextual relationships within narrative EMR data. These strategies collectively mitigate the inherent challenges associated with the preprocessing of narrative EMR data, resulting in a substantial performance improvement for HAE detection models.

4.3. Major findings and future works

This study presents a comprehensive scoping review of the preprocessing of narrative EMR to detect HAE. The initial step in conducting this scoping review adhered to the PRISMA-ScR guidelines, accompanied by the completion of the PRISMA checklist for the study protocol (see Appendix B line 280). The article selection process was conducted independently by two reviewers. Furthermore, two additional expert reviewers assessed the quality of the selected articles. Only those papers that achieved consensus among all four reviewers were incorporated into the full-text review. Four research questions were formulated, and comprehensive responses to each are provided.

Of the 48 studies, only 12 performed experiments that evaluated the impact of preprocessing techniques on HAE detection (including [31,33,38,39,42,53,55,56,58,64,69,73]). From these studies, a total of 114 experiments were extracted, and each was assigned an evaluation number to facilitate comparison. This comparison is valid only when experiments are conducted under the same HAE type, using the same data source, within the same institutional setting, and employing the same HAE detection approach.

The findings drawn from this study indicate that:

- **Finding research gaps:** The authors delineate three primary research gaps: (1) the imperative need for a pipeline to preprocess narrative EMR for the identification of HAE, (2) the necessity for a robust system capable of managing the extensive volume of narrative EMR data, and (3) the requirement for temporal event system, which are essential for effective HAE detection.
- **Impact of employing NLP rather than ICD codes in detecting HAE:** The study demonstrates that HAE detection utilizing narrative text in EMR data, when supplemented by NLP and ML techniques, exhibits superior performance compared to traditional HAE detection methods reliant on ICD codes or administrative data. Eight pieces of evidence from sixteen comparisons of experiments conducted under identical conditions is highlighted, specifically in works by [53,55].
- **Impact of handling large-scale data in detecting HAE:** Earlier approaches addressed the challenge of high data volume through section detection, sentence segmentation, and targeted keyword searches (e.g. [30,31,57,67,69]). Recent LLM-based studies leveraged prompting, chunking, and contextual reasoning (e.g., COT) to efficiently process extensive unstructured clinical narratives without the need for manual segmentation or heuristics (e.g. [42,45,46,49,65]). However, despite existing research, the approach for mitigating the substantial volume of data that significantly influences the performance of HAE detection remains a subject of ongoing investigation.
- **Impact of text normalization in detecting HAE:** Traditional systems necessitated extensive text normalization, which included word tokenization and normalization processes. The findings indicate that employing N-grams tokenization significantly enhances the efficacy of HAE detection. This is supported by eight pieces of evidence, as referenced in sources [33,53]. Additionally, eight pieces of evidence from [31,33] demonstrate that the integration of N-grams tokenization with various normalization techniques, such as lemmatization or synonym identification, significantly improves HAE detection performance. Conversely, LLM-based frameworks inherently incorporate tokenization and normalization within their architecture, thus rendering explicit normalization processes redundant. Nevertheless, questions persist in the LLM-based research context regarding whether the replacement of tailored tokenization and normalization methods — customized for clinical narrative text — with the default normalization and tokenization approaches of LLMs notably impacts the effectiveness of HAE detection.

- **Impact of feature engineering in detecting HAE:** A collective assessment of 33 pieces of evidence supports the conclusion that semantic feature extraction markedly enhances the performance of HAE detection, with notable contributions from 30 pieces of evidence cited in [58], including nine presented in Table 6, along with three additional evidences in [55,73]. When employing an LLM-based approach, the selection of inputs based on keywords or chunks that are closely related to the specific type of HAE significantly enhances the efficacy of HAE detection. This assertion is supported by six pieces of evidence cited in [42] and two pieces of evidence referenced in [64]. In the context of applications utilizing LLMs, the implementation of keywords and semantic chunks — sequences of words that exhibit significant semantic similarity to the type of HAE targeted for detection (for instance, "lower extremity thrombosis" and "upper extremity thrombosis" in the context of identifying DVT) — has proven to enhance the efficacy of HAE detection with LLMs. However, a crucial question remains regarding which of the employed approaches — chunking, prompting, COT, or keyword search — has the most significant impact on the performance of HAE detection.

The study underscores the significance of (1) employing N-gram extraction, (2) implementing stemming and/or lemmatization as techniques of word normalization, and (3) semantic feature extraction in feature engineering as essential preprocessing steps in the narrative EMR for enhancing the performance of HAE detection. While LLM-based systems naturally incorporate tokenization and normalization processes within their frameworks, it remains crucial to address features that hold semantic relevance to the specific type of HAE during preprocessing.

Future research should concentrate on addressing three unresolved questions in the area of HAE detection. First, despite the contributions of existing studies, uncertainty remains regarding which method of reducing the substantial volume of narrative text within EMR data yields the most significant performance enhancements in HAE detection. Second, within the realm of LLM-based methodologies, it is essential to examine whether substituting customized clinical text tokenization and normalization techniques with default tokenization and normalization approaches in LLMs has a notable impact on detection efficacy. Finally, an important area for further investigation is identifying which strategies — such as chunking, prompting, COT, or keyword search — most effectively enhance HAE detection outcomes.

4.4. Limitations of this work

The authors acknowledge several limitations associated with this study. Firstly, the research does not classify preprocessing into a specific category of HAE. The search strategy emphasizes keywords such as "hospital adverse event" and/or "adverse event", "medical error", "adverse effect", and/or "patient safety". While these keywords encompass various adverse events, there are no targeted keyword searches for each specific HAE (e.g., VTE and its synonyms, SSI and its synonyms, etc.). Secondly, the study does not investigate the interplay between HAE detection methods and preprocessing steps, nor does it assess their impact on the efficacy of HAE detection systems. For instance, Hinz et al. identified VTE utilizing a combination of NLP techniques and keyword searches without implementing text normalization. However, they concentrated on IE as part of feature engineering using the UMLS, achieving satisfactory performance [36]. Thirdly, this study gives limited attention to vectorization techniques within feature engineering, placing greater emphasis on IE techniques. The debate continues regarding which representation model delivers optimal performance. Although Sanyal et al. illustrated that Word-to-Vector exceeds the performance of TF–IDF and BioBERT [69], other investigations do not consistently affirm the superiority or advantages of one feature representation model over another (e.g., [63,72]). Fourthly, the study does not compare the effectiveness of various NLP techniques for text normalization, such as whether N-grams tokenization is more effective than lemmatization, or vice versa. Additionally, it does not evaluate whether text normalization as a preprocessing step is more effective than feature engineering techniques in detecting HAE. Lastly, while this study encompassed research that utilized LLM techniques for the detection of HAE, it did not examine the impact of individual IE techniques within LLMs, such as prompting, chunking, and COT reasoning, on the detection of HAE.

5. Conclusion

This study conducted a scoping review to examine the preprocessing of narrative EMR in order to identify HAE. The review yielded two principal findings. Firstly, it identified three research gaps: the necessity for a comprehensive preprocessing pipeline for narrative EMR, the requirement for systems capable of managing large data volumes, and the imperative for effective mechanisms for temporal event detection relevant to HAE identification. Addressing these gaps was the primary objective of the scoping review. Secondly, the study's secondary outcome emphasized the critical importance of preprocessing tasks in improving the accuracy of HAE detection. In particular, it underscored the significance of extracting N-grams, normalizing these N-grams through lemmatization and/or stemming, and considering semantic feature extraction during the preprocessing of narrative EMR. Future research should focus on addressing the identified research gaps and investigating the impact of preprocessing tasks in LLM-based models to enhance the performance of HAE detection.

CRediT authorship contribution statement

Hamed Jafarpour: Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization. **Guosong Wu:** Writing – review & editing, Methodology, Conceptualization. **Cheligeer (Ken) Cheligeer:** Writing – review & editing, Data curation. **Jun Yan:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Yuan Xu:** Writing – review & editing, Conceptualization. **Danielle A. Southern:** Writing – review & editing. **Cathy A. Eastwood:** Writing – review & editing. **Yong Zeng:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Hude Quan:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hude Quan's report received funding from the Canadian Institutes of Health Research. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This scoping review is funded by the Canadian Institutes of Health Research (CIHR).

Appendix A. PRISMA checklist

The PRISMA-ScR checklist is included with this article. This checklist delineates each section along with its corresponding page number where the information can be found.

Appendix B. Study protocol

The protocol for the scoping review study is provided as a PDF attachment for the reference. In accordance with the guidelines provided by Peters et al. in [17], the protocol is numbered sequentially from 1 to 449. Additionally, the study protocol checklist is available within the scoping review protocol, specifically located in lines 280 to 285. The study protocol has been included as an attachment to this review.

Appendix C. Supplementary data

All data extracted from the full-text review articles have been systematically organized into a table within a Microsoft Excel file, which is provided as supplementary data. Each article is assigned its own separate sheet in the file, identified by a unique "Study ID". Furthermore, the raw data extraction form can be found in Appendix B of the scoping review protocol, specifically starting from line 448.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.artmed.2025.103281.

References

[1] De Vries EN, Ramrattan MA, Smorenburg SM, Gouma DJ, Boermeester MA. The incidence and nature of in-hospital adverse events: a systematic review. BMJ Qual Saf 2008;17(3):216–23.

[2] Slawomirski L, Auraaen A, Klazinga NS. The economics of patient safety. OECD health working papers, OECD; 2017, URL https://www.oecd-ilibrary.org/content/paper/5a9858cd-en.

[3] Bates DW, Levine D, Syrowatka A, Kuznetsova M, Craig KJT, Rui A, Jackson GP, Rhee K. The potential of artificial intelligence to improve patient safety: a scoping review. NPJ Digit Med 2021;4(1):54.

[4] Klein DO, Rennenberg RJ, Koopmans RP, Prins MH. A systematic review of methods for medical record analysis to detect adverse events in hospitalized patients. J Patient Saf 2021;17(8):e1234–40.

[5] Estrada-Orozco K, Cruz FC, Cruz JB, Ruiz-Cardozo MA, Suárez-Chacón AM, Tribaldos JAC, Rojas MAC, Contreras RAR, González-Camargo JE, Berdugo JCG, et al. Hospital adverse event reporting systems: A systematic scoping review of qualitative and quantitative evidence. J Patient Saf 2021;17(8):e1866–72.

[6] Schwendimann R, Blatter C, Dhaini S, Simon M, Ausserhofer D. The occurrence, types, consequences and preventability of in-hospital adverse events–a scoping review. BMC Health Serv Res 2018;18:1–13.

[7] Meystre SM, Savova GK, Kipper-Schuler KC, Hurdle JF. Extracting information from textual documents in the electronic health record: a review of recent research. Yearb Med Inform 2008;17(01):128–44.

[8] Halpin H, Shortell SM, Milstein A, Vanneman M. Hospital adoption of automated surveillance technology and the implementation of infection prevention and control programs. Am J Infect Control 2011;39(4):270–6.

[9] van Mourik MS, Troelstra A, van Solinge WW, Moons KG, Bonten MJ. Automated surveillance for healthcare-associated infections: opportunities for improvement. Clin Infect Dis 2013;57(1):85–93.

[10] Govindan M, Van Citters AD, Nelson EC, Kelly-Cummings J, Suresh G. Automated detection of harm in healthcare with information technology: a systematic review. Qual Saf Heal Care 2010;19(5):e11.

[11] Goto M, Ohl ME, Schweizer ML, Perencevich EN. Accuracy of administrative code data for the surveillance of healthcare-associated infections: a systematic review and meta-analysis. Clin Infect Dis 2014;58(5):688–96.

[12] Classen DC, Resar R, Griffin F, Federico F, Frankel T, Kimmel N, Whittington JC, Frankel A, Seger A, James BC. 'Global trigger tool'shows that adverse events in hospitals may be ten times greater than previously measured. Health Aff 2011;30(4):581–9.

[13] Freeman R, Moore LS, Álvarez LG, Charlett A, Holmes A. Advances in electronic surveillance for healthcare-associated infections in the 21st century: a systematic review. J Hosp Infect 2013;84(2):106–19.

[14] Groene O, Kristensen S, Arah O, Thompson C, Bartels P, Sunol R, Klazinga N, Consortium DP, Klazinga N, Kringos D, et al. Feasibility of using administrative data to compare hospital performance in the EU. Int J Qual Heal Care 2014;26(suppl_1):108–15.

[15] Schuantes-Paim SM, Wachholz LF, Knihs NDS, De Oliveira PC, Roza BDA, Schirmer J. Adverse events reporting systems in cells, organs, and tissues donation and transplantation: scoping review. Transplant Proc 2023;55(6):1352–8.

[16] Bhangu G, Murray A, Qayyum A, Goumeniouk N, Goodacre S, Hunt BJ, Touhami O, Tester J, Rees M, Hammerschlag G, et al. Diagnostic strategies in postpartum individuals with suspected venous thromboembolism: A scoping review. Thromb Res 2024.

[17] Peters MD, Marnie C, Tricco AC, Pollock D, Munn Z, Alexander L, McInerney P, Godfrey CM, Khalil H. Updated methodological guidance for the conduct of scoping reviews. JBI Evid Synth 2020;18(10):2119–26.

[18] Tricco AC, Lillie E, Zarin W, O'brien K, Colquhoun H, Kastner M, Levac D, Ng C, Sharpe JP, Wilson K, et al. A scoping review on the conduct and reporting of scoping reviews. BMC Med Res Methodol 2016;16:1–10.

[19] Daniel Jurafsky JHM. Speech and language processing. 2024, https://web.stanford.edu/~jurafsky/slp3/2.pdf. [Accessed 20 June 2024].

[20] Peters MD, Godfrey C, McInerney P, Munn Z, Tricco AC, Khalil H, et al. Chapter 11: scoping reviews. JBI Man Evid Synth 2020;169(7):467–73.

[21] Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, Moher D, Peters MD, Horsley T, Weeks L, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 2018;169(7):467–73.

[22] Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. Bmj 2015;349.

[23] PROSPERO. International prospective register of systematic reviews. https://www.crd.york.ac.uk/prospero/, [Accessed 16 April 2024].

[24] OSF. Open Science Framework. https://osf.io/, [Accessed 16 April 2024].

[25] Figshare. Figshare. https://figshare.com/, [Accessed 16 April 2024].

[26] Melton GB, Hripcsak G. Automated detection of adverse events using natural language processing of discharge summaries. J Am Med Inform Assoc 2005;12(4):448–57.

[27] Viera AJ, Garrett JM, et al. Understanding interobserver agreement: the kappa statistic. Fam Med 2005;37(5):360–3.

[28] Tian Z, Sun S, Eguale T, Rochefort CM. Automated extraction of VTE events from narrative radiology reports in electronic health records: a validation study. Med Care 2017;55(10):e73–80.

[29] Woller B, Daw A, Aston V, Lloyd J, Snow G, Stevens SM, Woller SC, Jones P, Bledsoe J. Natural language processing performance for the identification of venous thromboembolism in an integrated healthcare system. Clin Appl Thromb Hemost 2021;27:10760296211013108.

[30] Verma AA, Masoom H, Pou-Prom C, Shin S, Guerzhoy M, Fralick M, Mamdani M, Razak F. Developing and validating natural language processing algorithms for radiology reports compared to ICD-10 codes for identifying venous thromboembolism in hospitalized medical patients. Thromb Res 2022;209:51–8.

[31] Shi J, Hurdle JF, Johnson SA, Ferraro JP, Skarda DE, Finlayson SR, Samore MH, Bucher BT. Natural language processing for the surveillance of postoperative venous thromboembolism. Surgery 2021;170(4):1175–82.

[32] Dantes RB, Zheng S, Lu JJ, Beckman MG, Krishnaswamy A, Richardson LC, Chernetsky-Tejedor S, Wang F. Improved identification of venous thromboembolism from electronic medical records using a novel information extraction software platform. Med Care 2018;56(9):e54–60.

[33] Rochefort CM, Verma AD, Eguale T, Lee TC, Buckeridge DL. A novel method of adverse event detection can accurately identify venous thromboembolisms (VTEs) from narrative electronic health record data. J Am Med Inform Assoc 2015;22(1):155–65.

[34] Muñoz AJ, Souto JC, Lecumberri R, Obispo B, Sanchez A, Aparicio J, Aguayo C, Gutierrez D, Palomo AG, Fanjul V, et al. Development of a predictive model of venous thromboembolism recurrence in anticoagulated cancer patients using machine learning. Thromb Res 2023;228:181–8.

[35] Murff HJ, FitzHenry F, Matheny ME, Gentry N, Kotter KL, Crimin K, Dittus RS, Rosen AK, Elkin PL, Brown SH, et al. Automated identification of postoperative complications within an electronic medical record using natural language processing. Jama 2011;306(8):848–55.

[36] Hinz ERM, Bastarache L, Denny JC. A natural language processing algorithm to define a venous thromboembolism phenotype. In: AMIA annual symposium proceedings. vol. 2013, American Medical Informatics Association; 2013, p. 975.

[37] Gálvez JA, Pappas JM, Ahumada L, Martin JN, Simpao AF, Rehman MA, Witmer C. The use of natural language processing on pediatric diagnostic radiology reports in the electronic health record to identify deep venous thrombosis in children. J Thromb Thrombolysis 2017;44:281–90.

[38] Wang J, de Vale JS, Gupta S, Upadhyaya P, Lisboa FA, Schobel SA, Elster EA, Dente CJ, Buchman TG, Kamaleswaran R. ClotCatcher: a novel natural language model to accurately adjudicate venous thromboembolism from radiology reports. BMC Med Inform Decis Mak 2023;23(1):262.

[39] Deng J, Wu Y, Hayssen H, Englum B, Kankaria A, Mayorga-Carlin M, Sahoo S, Sorkin J, Lal B, Yesha Y, et al. Improving vte identification through adaptive nlp model selection and clinical expert rule-based classifier from radiology reports. In: 2023 IEEE international conference on bioinformatics and biomedicine. BIBM, IEEE; 2023, p. 480–7.

[40] Martín AJM, Lecumberri R, Souto JC, Obispo B, Sanchez A, Aparicio J, Aguayo C, Gutierrez D, Palomo AG, Benavent D, et al. Prediction model for major bleeding in anticoagulated patients with cancer-associated venous thromboembolism using machine learning and natural language processing. Clin Transl Oncol 2024;27(4):1816.

[41] Jin Z-G, Zhang H, Tai M-H, Yang Y, Yao Y, Guo Y-T. Natural language processing in a clinical decision support system for the identification of venous thromboembolism: algorithm development and validation. J Med Internet Res 2023;25:e43153.

[42] Cheligeer C, Southern DA, Yan J, Wu G, Pan J, Lee S, Martin EA, Jafarpour H, Eastwood CA, Zeng Y, et al. Utilizing large language models for detecting hospital-acquired conditions: an empirical study on pulmonary embolism. J Am Med Inform Assoc 2025;32(5):876–84.

[43] Piscitelli A, Bevilacqua L, Labella B, Parravicini E, Auxilia F. A keyword approach to identify adverse events within narrative documents from 4 Italian institutions. J Patient Saf 2022;18(1):e362–7.

[44] da Silva DA, Ten Caten CS, Dos Santos RP, Fogliatto FS, Hsuan J. Predicting the occurrence of surgical site infections using text mining and machine learning. PloS One 2019;14(12):e0226272.

[45] Hsueh JY, Nethala D, Singh S, Linehan WM, Ball MW. Investigating the clinical reasoning abilities of large language model GPT-4: an analysis of postoperative complications from renal surgeries. In: Urologic oncology: seminars and original investigations. vol. 42, Elsevier; 2024, p. 292–e1.

[46] Shaheen A, Afflitto GG, Swaminathan SS. ChatGPT-assisted classification of post-operative bleeding following microinvasive glaucoma surgery using electronic health record data. Ophthalmol Sci 2025;5(1):100602.

[47] Dencker EE, Bonde A, Troelsen A, Sillesen M. Assessing the utility of natural language processing for detecting postoperative complications from free medical text. BJS Open 2024;8(2):zrae020.

[48] Biswas S, McMenemy L, Sarkar V, MacArthur J, Snowdon E, Tetlow C, George KJ. Natural language processing for the automated detection of intra-operative elements in lumbar spine surgery. Front Surg 2023;10:1271775.

[49] Zhu H, Wang R, Qian J, Wu Y, Jin Z, Shan X, Ji F, Yuan Z, Pan T. Leveraging large language models for predicting postoperative acute kidney injury in elderly patients. BME Front 2025;6:0111.

[50] Badia JM, Casanova-Portoles D, Membrilla E, Rubiés C, Pujol M, Sancho J. Evaluation of ChatGPT-4 for the detection of surgical site infections from electronic health records after colorectal surgery: A pilot diagnostic accuracy study. J Infect Public Heal 2025;18(2):102627.

[51] Xu P. Multi-layered data framework for enhancing postoperative outcomes and anaesthesia management through natural language processing. SLAS Technol 2025;100294.

[52] Weller GB, Lovely J, Larson DW, Earnshaw BA, Huebner M. Leveraging electronic health records for predictive modeling of post-surgical complications. Stat Methods Med Res 2018;27(11):3271–85.

[53] Thirukumaran CP, Zaman A, Rubery PT, Calabria C, Li Y, Ricciardi BF, Bakhsh WR, Kautz H. Natural language processing for the identification of surgical site infections in orthopaedics. JBJS 2019;101(24):2167–74.

[54] Shen F, Larson DW, Naessens JM, Habermann EB, Liu H, Sohn S. Detection of surgical site infection utilizing automated feature generation in clinical notes. J Heal Inform Res 2019;3:267–82.

[55] Chapman AB, Mowery DL, Swords DS, Chapman WW, Bucher BT. Detecting evidence of intra-abdominal surgical site infections from radiology reports using natural language processing. In: AMIA annual symposium proceedings. vol. 2017, American Medical Informatics Association; 2017, p. 515.

[56] Karhade AV, Bongers ME, Groot OQ, Cha TD, Doorly TP, Fogel HA, Hershman SH, Tobert DG, Schoenfeld AJ, Kang JD, et al. Can natural language processing provide accurate, automated reporting of wound infection requiring reoperation after lumbar discectomy? Spine J 2020;20(10):1602–9.

[57] Shiner B, Neily J, Mills PD, Watts BV. Identification of inpatient falls using automated review of text-based medical records. J Patient Saf 2020;16(3):e174–8.

[58] Wang Y, Coiera E, Magrabi F. Can unified medical language system–based semantic representation improve automated identification of patient safety incident reports by type and severity? J Am Med Inform Assoc 2020;27(10):1502–9.

[59] Cho I, Park H, Park BS, Lee D-g. Enhancing adverse event reporting with clinical language models: Inpatient falls. J Adv Nurs 2025.

[60] Wasylewicz A, van de Burgt B, Weterings A, Jessurun N, Korsten E, Egberts T, Bouwman A, Kerskes M, Grouls R, van der Linden C. Identifying adverse drug reactions from free-text electronic hospital health record notes. Br J Clin Pharmacol 2022;88(3):1235–45.

[61] Chopard D, Treder MS, Corcoran P, Ahmed N, Johnson C, Busse M, Spasic I, et al. Text mining of adverse events in clinical trials: deep learning approach. JMIR Med Inform 2021;9(12):e28632.

[62] Gupta S, Belouali A, Shah N, Atkins M, Madhavan S. Identification of patients with immune-related adverse events from clinical notes using machine learning. In: 2020 IEEE international conference on healthcare informatics. ICHI, IEEE; 2020, p. 1–2.

[63] Gupta S, Belouali A, Shah NJ, Atkins MB, Madhavan S. Automated identification of patients with immune-related adverse events from clinical notes using word embedding and machine learning. JCO Clin Cancer Inform 2021;5:541–9.

[64] Shashikumar SP, Mohammadi S, Krishnamoorthy R, Patel A, Wardi G, Ahn JC, Singh K, Aronoff-Spencer E, Nemati S. Development and prospective implementation of a large language model based system for early sepsis prediction. Npj Digit Med 2025;8(1):1–10.

[65] Zhang W, Wu M, Zhou L, Shao M, Wang C, Wang Y. A sepsis diagnosis method based on chain-of-thought reasoning using large language models. Biocybern Biomed Eng 2025;45(2):269–77.

[66] Mertes PM, Morgand C, Barach P, Jurkolow G, Assmann KE, Dufetelle E, Susplugas V, Alauddin B, Yavordios PG, Tourres J, et al. Validation of a natural language processing algorithm using national reporting data to improve identification of anesthesia-related ADVerse events: the "ADVENTURE" study. Anaesth Crit Care Pain Med 2024;43(4):101390.

[67] Mao J, Sedrakyan A, Sun T, Guiahi M, Chudnoff S, Kinard M, Johnson SB. Assessing adverse event reports of hysteroscopic sterilization device removal using natural language processing. Pharmacoepidemiol Drug Safety 2022;31(4):442–51.

[68] Ozonoff A, Milliren CE, Fournier K, Welcher J, Landschaft A, Samnaliev M, Saluvan M, Waltzman M, Kimia AA. Electronic surveillance of patient safety events using natural language processing. Heal Inform J 2022;28(4):14604582221132429.

[69] Sanyal J, Rubin D, Banerjee I. A weakly supervised model for the automated detection of adverse events using clinical notes. J Biomed Inform 2022;126:103969.

[70] Rochefort CM, Buckeridge DL, Tanguay A, Biron A, D'Aragon F, Wang S, Gallix B, Valiquette L, Audet L-A, Lee TC, et al. Accuracy and generalizability of using automated methods for identifying adverse events from electronic health record data: a validation study protocol. BMC Health Serv Res 2017;17:1–9.

[71] Whitaker B, Pizarro J, Deady M, Williams A, Ezzeldin H, Belov A, Kanderian S, Billings D, Cook K, Hettinger AZ, et al. Detection of allergic transfusion-related adverse events from electronic medical records. Transfusion 2022;62(10):2029–38.

[72] Borjali A, Magnéli M, Shin D, Malchau H, Muratoglu OK, Varadarajan KM. Natural language processing with deep learning for medical adverse event detection from free-text medical narratives: A case study of detecting total hip replacement dislocation. Comput Biol Med 2021;129:104140.

[73] Sabra S, Malik KM, Alobaidi M. Prediction of venous thromboembolism using semantic and sentiment analyses of clinical narratives. Comput Biol Med 2018;94:1–10.

[74] Mohammadi R, Jain S, Namin AT, Scholem Heller M, Palacholla R, Kamarthi S, Wallace B. Predicting unplanned readmissions following a hip or knee arthroplasty: retrospective observational study. JMIR Med Inform 2020;8(11):e19761.

[75] Zheng S, Lu JJ, Ghasemzadeh N, Hayek SS, Quyyumi AA, Wang F, et al. Effective information extraction framework for heterogeneous clinical reports using online machine learning and controlled vocabularies. JMIR Med Inform 2017;5(2):e7235.

[76] Wang Y, Wang L, Rastegar-Mojarad M, Moon S, Shen F, Afzal N, Liu S, Zeng Y, Mehrabi S, Sohn S, et al. Clinical information extraction applications: a literature review. J Biomed Inform 2018;77:34–49.

[77] Navarro DF, Ijaz K, Rezazadegan D, Rahimi-Ardabili H, Dras M, Coiera E, Berkovsky S. Clinical named entity recognition and relation extraction using natural language processing of medical free text: A systematic review. Int J Med Inform 2023;177:105122.

[78] Fu S, Chen D, He H, Liu S, Moon S, Peterson KJ, Shen F, Wang L, Wang Y, Wen A, et al. Clinical concept extraction: a methodology review. J Biomed Inform 2020;109:103526.

[79] Zheng A, Casari A. Feature engineering for machine learning: principles and techniques for data scientists. " O'Reilly Media, Inc."; 2018.

[80] Friedman C. A broad-coverage natural language processing system. In: Proceedings of the AMIA symposium. American Medical Informatics Association; 2000, p. 270.

[81] Chapman BE, Lee S, Kang HP, Chapman WW. Document-level classification of CT pulmonary angiography reports based on an extension of the ConText algorithm. J Biomed Inform 2011;44(5):728–37.

[82] Penz JF, Wilcox AB, Hurdle JF. Automated identification of adverse events related to central venous catheters. J Biomed Inform 2007;40(2):174–82.

[83] Bird S, Klein E, Loper E. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc."; 2009.

[84] spaCy. spaCy 3: Industrial-Strength Natural Language Processing in Python. https://spacy.io, [Accessed 16 April 2024].

[85] Qi P, Zhang Y, Zhang Y, Bolton J, Manning CD. Stanza: A Python natural language processing toolkit for many human languages. 2020, arXiv preprint arXiv:2003.07082.

[86] Chapman WW, Bridewell W, Hanbury P, Cooper GF, Buchanan BG. A simple algorithm for identifying negated findings and diseases in discharge summaries. J Biomed Inform 2001;34(5):301–10.

[87] Savova GK, Masanz JJ, Ogren PV, Zheng J, Sohn S, Kipper-Schuler KC, Chute CG. Mayo clinical text analysis and knowledge extraction system (CTAKES): architecture, component evaluation and applications. J Am Med Inform Assoc 2010;17(5):507–13.

[88] Eyre H, Chapman AB, Peterson KS, Shi J, Alba PR, Jones MM, Box TL, DuVall SL, Patterson OV. Launching into clinical space with medspaCy: a new clinical text processing toolkit in Python. In: AMIA annual symposium proceedings. vol. 2021, American Medical Informatics Association; 2021, p. 438.

[89] Zhang Y, Zhang Y, Qi P, Manning CD, Langlotz CP. Biomedical and clinical English model packages for the Stanza Python NLP library. J Am Med Inform Assoc 2021;28(9):1892–9.

[90] Neumann M, King D, Beltagy I, Ammar W. ScispaCy: fast and robust models for biomedical natural language processing. 2019, arXiv preprint arXiv:1902.07669.
[91] Wang S, Lin M, Ding Y, Shih G, Lu Z, Peng Y. Radiology text analysis system (RadText): architecture and evaluation. In: 2022 IEEE 10th international conference on healthcare informatics. ICHI, IEEE; 2022, p. 288–96.
[92] Friedman C, Shagina L, Lussier Y, Hripcsak G. Automated encoding of clinical documents based on natural language processing. J Am Med Inform Assoc 2004;11(5):392–402.
[93] Joshi M, Mecklai K, Rozenblum R, Samal L. Implementation approaches and barriers for rule-based and machine learning-based sepsis risk prediction tools: a qualitative study. JAMIA Open 2022;5(2):ooac022.
[94] Tayefi M, Ngo P, Chomutare T, Dalianis H, Salvi E, Budrionis A, Godtliebsen F. Challenges and opportunities beyond structured data in analysis of electronic health records. Wiley Interdiscip Rev: Comput Stat 2021;13(6):e1549.
[95] Martínez-García M, Hernández-Lemus E. Data integration challenges for machine learning in precision medicine. Front Med 2022;8:784455.
[96] Zuckerman RB, Sheingold SH, Orav EJ, Ruhter J, Epstein AM. Readmissions, observation, and the hospital readmissions reduction program. N Engl J Med 2016;374(16):1543–51.
[97] Sundararaman A, Ramanathan SV, Thati R. Novel approach to predict hospital readmissions using feature selection from unstructured data with class imbalance. Big Data Res 2018;13:65–75.
[98] Donnellan E, Khorana AA. Cancer and venous thromboembolic disease: a review. Oncol 2017;22(2):199–207.
[99] Horsted F, West J, Grainge MJ. Risk of venous thromboembolism in patients with cancer: a systematic review and meta-analysis. PLoS Med 2012;9(7).